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AGformer: Adaptive filtering and gated fusion-attention for multivariate time series forecasting

Shu-Xin Xu [a,b], Jiang-Wen Xiao [a,b], Peng-Cheng Li [a,b], Tian-Hao Lu [a,b,*]

[a] School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, 430074, China
[b] Key Laboratory of Image Processing and Intelligent Control (Huazhong University of Science and Technology), Ministry of Education, Wuhan, 430074, China

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ABSTRACT

Multivariate time series forecasting is a fundamental problem in sequence modeling, with broad applications spanning energy systems, meteorology, finance, and transportation. Despite significant progress in Transformer-based, TCN-based, and MLP-based architectures, three persistent neural network design challenges remain unresolved: 1) **non-adaptive frequency-domain filtering** that applies identical spectral parameters across heterogeneous variables; 2) **inadequate utilization of informative short-term fluctuations** that are discarded as noise yet encode critical event-driven dynamics; and 3) **insufficient exploration of inter-scale dependencies** in multi-scale temporal representations. To address these challenges, this paper proposes **AGformer**, a novel forecasting architecture integrating three complementary neural modules: 1) A **variable-oriented Adaptive Multi-k Filter (AKF)** that dynamically generates per-variable spectral filtering parameters from learned frequency-domain features, eliminating the need for manual global hyperparameter tuning. 2) A **Dual-Gate Fusion-Attention (DGFA) mechanism** that synergistically models long-term trends and short-term fluctuations via complementary temporal and content gating, recovering event-driven signals that are otherwise discarded by standard denoising. 3) A **Multi-Scale Embedding (MSED) layer** that fuses temporal patterns across resolutions through attention-based inter-scale weighting, capturing cross-resolution hierarchical dependencies that simple concatenation or summation cannot model. Extensive experiments on nine representative benchmark datasets across electricity, weather, finance, and transportation domains demonstrate AGformer's consistent superiority, achieving average reductions of 8.1%–23.6% in MSE and MAE relative to state-of-the-art baselines across nine benchmarks.

1. Introduction

Multivariate time series forecasting is a core problem in sequence modeling, with broad applications spanning energy systems, meteorology, finance, and transportation (Rahman et al., 2025). Real-world multivariate signals are inherently heterogeneous: individual channels exhibit contrasting spectral profiles, abrupt short-term events coexist with smooth long-term trends, and informative patterns emerge simultaneously across multiple temporal scales (Liu et al., 2025a; Pentsos et al., 2025). These properties pose fundamental challenges for neural sequence models. Recent advances in time series forecasting can be broadly categorized into two paradigms: (i) **specialized small models** trained from scratch for specific domains, and (ii) **time-series foundation models** (TSFMs) pre-trained on large-scale cross-domain corpora (Das et al., 2024; Goswami et al., 2024; Jin et al., 2024). This paper focuses on the former, targeting scenarios where efficiency and low-latency inference are prioritized. Within specialized small models, three mainstream methodological streams exist, each suffering from a persistent limitation:

1. Frequency-domain methods with non-adaptive filtering. Models like FEDformer (Zhou et al., 2022) and SDformer (Zhou et al., 2024) process time series in the frequency domain for denoising. While effective, they rely on manually tuned global parameters, such as a fixed number of TopK components or random selection, which are applied uniformly across all variables. This lack of adaptability means that smooth variables may retain excessive noise, while volatile ones may lose informative high-frequency components. From a neural network design perspective, the absence of learnable, end-to-end parameterization for spectral filtering prevents the model from acquiring channel-specific inductive biases during training, forcing suboptimal generalization across heterogeneous variables.

* Corresponding author at: School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan, 430074, China.
E-mail address: D202381397@hust.edu.cn (T.-H. Lu).

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2. Long- and short-term modeling methods that discard informative fluctuations. Many models inadvertently remove short-term fluctuations during preprocessing or denoising, treating them as mere noise. For instance, the Non-stationary Transformer (Liu et al., 2022) employs series smoothing, which can over-smooth critical spikes caused by abrupt structural breaks or sudden load transitions. This inadequate utilization of short-term signals limits model robustness in real-world scenarios where such "noise" encodes critical information about system state changes. Architecturally, existing attention mechanisms lack a dedicated gating pathway to selectively retrieve suppressed high-frequency signals, resulting in an information bottleneck.
3. Multi-scale modeling with simplistic fusion.Models like TimesNet (Wu et al., 2022) and Autoformer (Wu et al., 2021) capture temporal patterns at different scales but fuse them using elementary operations such as concatenation (Zhou et al., 2020) or weighted summation (Wu et al., 2021). This insufficient exploration of inter-scale dependencies fails to model the complex, dynamic interactions between short-term granularity and long-term trends. Static fusion operators cannot adaptively reweight inter-scale contributions conditioned on input content, limiting the depth of cross-resolution feature interaction.

Crucially, these three limitations are *coupled*: stronger denoising (to address limitation 1) directly worsens the loss of fluctuations (limitation 2); multi-scale pooling (limitation 3) further smooths the signal, also exacerbating limitation 2. Isolated improvements within any single methodological stream have thus reached diminishing returns. To break this deadlock, we propose **AGformer**, a specialized architecture that jointly addresses all three bottlenecks through three tightly coupled modules:

1. We propose a **variable-oriented Adaptive Filter (AKF)** that leverages multiple frequency-domain features (such as average energy and spectral entropy) to learn variable-specific spectral sparsity, thereby replacing manually set global TopK thresholds. Unlike existing learnable spectral filters that learn shared continuous coefficients, AKF learns discrete sparsity levels for each channel; this fundamentally shifts the optimization objective from "how much to attenuate" to "how many frequency components to retain."
2. We design a **Dual-Gate Fusion-Attention (DGFA) mechanism** that explicitly models the interaction between short-term fluctuation information and long-term trends, utilizing complementary temporal and content gating to selectively focus on fluctuation signals. Unlike standard self-attention that operates on a single information stream or GLU that applies gating within feedforward layers, DGFA is a cross-attention mechanism that bridges two distinct streams: raw data (containing fluctuations) as queries, and filtered trends as keys and values.
3. We develop a **Multi-Scale Embedding (MSED) layer** that employs an attention-based fusion strategy to extract and fuse temporal patterns across different scales, enabling dynamic cross-resolution interactions. Unlike simple concatenation, pooling, or static linear mixing that treat scales independently, MSED computes attention across the scale dimension itself—achieving explicit scale-to-scale interaction rather than within-scale processing.

These modules constitute a "closed-loop collaborative" mechanism: the AKF provides denoised trend information to the MSED; the MSED supplies multi-resolution keys and values to the DGFA; and the DGFA utilizes the raw data as queries to precisely focus on the fluctuation components previously filtered out by the AKF. This cyclic dependency ensures the synergistic optimization of filtering, multi-scale representation, and fluctuation recovery—distinguishing the model from previous approaches that relied on isolated, independently tuned modules. Although time-series foundation models (TSFM) such as Time-LLM (Jin et al., 2024) excel at cross-domain zero-shot generalization, they lack specialized mechanisms to handle variable heterogeneity and local fluctuations, and they entail significantly higher inference costs. In contrast, AGformer is a lightweight, specialized model offering explicit inductive biases for these critical capabilities.

Extensive experiments on nine benchmark datasets confirm that AGformer consistently outperforms existing state-of-the-art models, achieving significant improvements in both MSE and MAE. The remainder of this paper is organized as follows. Section 2 provides a detailed literature review and identifies gaps in current research. Section 3 elaborates the proposed model and its components. Section 4 presents the prediction results, ablation studies, and visualization outcomes. Section 5 concludes with key findings and implications.

2. Related work

2.1. Frequency-domain filtering

Frequency-domain processing provides a powerful tool for denoising and feature extraction in time series analysis. Pioneering models like FEDformer (Zhou et al., 2022) incorporated frequency enhancement strategies; however, their random selection of frequency components lacks a data-driven rationale. Subsequent works, such as SDformer (Zhou et al., 2024) and FEDAF (Yang et al., 2024), widely adopted TopK filtering strategy, which retains the K frequency components with the highest magnitudes. This frequency domain filtering strategy is selected because periodic patterns in the load data (e.g., daily, weekly) manifest as frequencies with concentrated energy, enabling effective separation of dominant modes from noise.

A key limitation of these methods is that the filtering parameter K is a global hyperparameter, requiring manual tuning, which overlooks the spectral differences between variables (e.g., base load vs. highly volatile industrial load), potentially oversmoothing some and underdenoising others. While MFformer (Tong & Liu, 2025) and TSLANet (Eldele et al., 2024) attempted to improve adaptability by identifying dominant frequencies via the power spectrum, they still rely on predefined algorithms or thresholds, lacking an end-to-end, learnable mechanism for variable-specific filtering. For instance, although JTFT (Chen et al., 2024) introduces learnable frequencies, it still relies on global DCT bases for initialization and lacks cross-channel adaptivity. TCDformer (Wan et al., 2024) utilizes wavelet transforms, where the frequency decomposition similarly depends on fixed wavelet bases.

The lack of adaptability motivates our design of the Adaptive Multi-k Filter (AKF) module, which dynamically generates variable-specific K values by learning from frequency-domain features, making filtering a self-adaptive capability of the model. Critically, unlike methods that learn shared filter coefficients or rely on fixed bases, AKF learns a discrete sparsity level per variable—a fundamentally different objective that enables per-channel spectral tailoring.

2.2. Long-term and short-term modeling

Accurate forecasting requires capturing long-term trends and responding to short-term fluctuations from unforeseen events, but many models discard these variations, treating them as noise during denoising or smoothing. For instance, the Non-stationary Transformer (Liu et al., 2022) employs series smoothing to handle non-stationarity, but this can suppress critical short-term spikes. Similarly, decomposition-reconstruction methods (Tong & Liu, 2025) often discard high-frequency components deemed "noise" during reconstruction. In contrast, TCDformer (Li et al., 2025) explicitly detects and preserves change-point information. MSCformer (Li et al., 2024a) designs a local dependency compensation network; and CTFformer (Fu et al., 2024) extracts short-term fluctuations via cross-convolution. These endeavors collectively underscore the importance of short-term signals. We model short-term fluctuations as they contain critical information about signal dynamics and non-stationary transitions, crucial for robust forecasting.

To capture long-term trends and short-term fluctuations, we adopt the **Transformer** as our foundational architecture, as numerous studies have demonstrated its capability to model global dependencies. Crossformer (Zhang & Yan, 2023) integrates long-term and short-term features through a hierarchical encoder-decoder, but its attention mechanism is confined to individual scales, lacking dynamic cross-scale interactions, which limits coordination between long-term and short-term information. iTransformer (Liu et al., 2023) processes each variable sequence independently, learning long-term trends and short-term details, but its attention mechanism only uncovers variable correlations, lacking direct capture of dynamic interactions between long- and short-term time points. Li et al. (2024b) improves long- and short-term dependency capture by augmenting iTransformer (Liu et al., 2023) with TVF-EMD decomposition and SAA, but the decomposition and reconstruction process adds complexity and still faces limitations in handling high-frequency fluctuations.

The idea of using gating to balance information flow has deep roots in the neural network literature. LSTM (Hochreiter & Schmidhuber, 1997) and GRU (Cho et al., 2014) demonstrated that learnable gates are essential for selectively retaining or discarding sequential information, a principle central to our DGFA design. More recently, Gated Linear Units (GLU) (Dauphin et al., 2017) extended this concept to feed-forward networks, showing that gating enables effective feature selection without recurrence.

To address these issues, we design DGFA as a cross-attention mechanism that dynamically bridges the filtered trend stream and the raw fluctuation stream. Unlike standard self-attention (single-stream) or GLU (FFN-level), DGFA applies dual gating within the cross-attention framework: TempGate captures local temporal windows via depthwise convolution, while ContentGate evaluates cross-feature semantic relevance, jointly enabling selective fusion of trend and fluctuation information across two distinct information streams.

2.3. Multi-scale fusion

Time series patterns span multiple time scales (e.g., hourly, daily, weekly), and modeling at a single scale overlooks fine-grained variations and long-term trend shifts (Li et al., 2019). Therefore, multi-scale modeling is necessary for a comprehensive understanding of temporal dynamics. TimesNet (Wu et al., 2022) effectively reveals intra-period and inter-period variations by transforming 1D time series into a set of 2D tensors based on different periods. However, its fusion of multi-scale features relies on simple concatenation or pooling operations. Likewise, Autoformer (Wu et al., 2021), despite its excellent multi-scale analysis achieved through series decomposition and an autocorrelation mechanism, employs a static weighted summation for fusion. Informer (Zhou et al., 2020) uses a sparse attention mechanism for computational efficiency and a multi-scale compression strategy, but it relies on simple splicing for fusion, limiting the depth of information exploration. Xiao et al. (2023) uses VMD decomposition and dual attention mechanisms for multi-scale separation and fusion of trends and cycles, but the fusion method relies on splicing operations and lacks deep interaction. LightTS (Zhang et al., 2022) captures local-global dependencies with continuous and interval sampling, but its MLP structure has limited expressive ability for complex multi-variable interactions. TimeMixer (Wang et al., 2024a) combines trend and seasonal information using a decomposable multi-scale mixing architecture, but its linear mixing structure has limited adaptability to complex patterns. Consequently, numerous representative studies on multi-scale fusion have emerged. MSA-LR (Sun et al., 2025) explicitly learns weight matrices for multiple adaptive time scales and utilizes low-rank approximation techniques to reduce computational complexity. The Dual-Dimensional Fusion (DuDF) mechanism incorporated within the DiffMixer (Zhang et al., 2025) model enables the simultaneous weighted fusion of different frequency components in both high-dimensional and original-dimensional spaces.

Given that simple fusion operations are unable to dynamically weigh the information contributed by different scales, we were inspired to refine existing multi-scale fusion strategies. The Multi-Scale Embedding (MSED) layer, integrated into the Transformer, uses an attention-based fusion strategy to capture complex dependencies across scales. Unlike concatenation, pooling, or linear mixing that treat scales as independent features, MSED computes attention across the scale dimension itself—each scale's representation is dynamically reweighted based on its relevance to a global prototype aggregated from all scales, achieving explicit scale-to-scale interaction.

The importance of multi-scale temporal feature extraction has also been established in convolutional neural network architectures. WaveNet (van den Oord et al., 2016) pioneered dilated causal convolutions to efficiently capture long-range dependencies across exponentially growing receptive fields. TCN (Bai et al., 2018) further demonstrated that temporal convolutional networks with hierarchical dilated convolutions can match or outperform recurrent architectures on sequence modeling tasks, leveraging multi-scale receptive fields without the vanishing gradient problem. While these architectures extract multi-scale features implicitly through stacked dilated convolutions, MSED makes inter-scale feature integration explicit and dynamic through an attention-based fusion mechanism, enabling the model to adaptively weight contributions from different temporal resolutions.

2.4. Large time-series models

In recent years, time-series foundation models (TSFMs) pre-trained on massive, cross-domain corpora have been rapidly emerging. Examples include Sundial (Liu et al., 2025b), a flow-matching-based generative foundation model with 128 million parameters—pre-trained on a dataset containing one trillion time points—that achieves millisecond-level zero-shot inference; and Chronos-2 (Ansari et al., 2025), an encoder-only model with 120 million parameters based on the T5 architecture, which supports zero-shot forecasting for univariate and multivariate series (including those with covariate information) via a group attention mechanism designed for in-context learning. Moirai 2.0, another representative large-scale decoder-only model, employs quantile forecasting and multi-token prediction techniques.

Although TSFM models demonstrate strong zero-shot generalization capabilities, they possess inherent limitations: their pre-training prioritizes cross-domain generalization over domain-specific fidelity; they lack explicit inductive biases addressing variable-specific spectral heterogeneity and local abrupt structural changes; and their inference requires substantial computational resources.

In contrast to TSFM models that emphasize cross-domain zero-shot generalization, AGformer is a compact model specifically designed for scenarios where domain-specific fidelity, efficiency, and interpretability are paramount. These two paradigms are complementary rather than competitive.

3. AGformer

3.1. Overall architecture

Consider a multivariate time series sample with a lookback window T: $\mathbf{X}_t = [\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_T] \in \mathbb{R}^{V \times T}$, where V is the number of variables. $\mathbf{X}_t$ is the model input. Denote the output as: $\mathbf{Y}_t^* = [\mathbf{X}_{T+1}, \mathbf{X}_{T+2}, ..., \mathbf{X}_{T+\tau}] \in \mathbb{R}^{V \times \tau}$, where τ is the forecasting step of the model. Define a mapping function f: $\mathbb{R}^{V \times T} \to \mathbb{R}^{V \times \tau}$ as:

$$\mathbf{Y}_t^* = f(\mathbf{X}_t) \tag{1}$$

Fig. 1 illustrates the overall architecture of the proposed AGformer model. The Adaptive Multi-k Filter (AKF) denoises the model inputs, and the Multi-Scale Embedding (MSED) layer extracts patterns of different temporal scales for the filtered trend information and fuses them

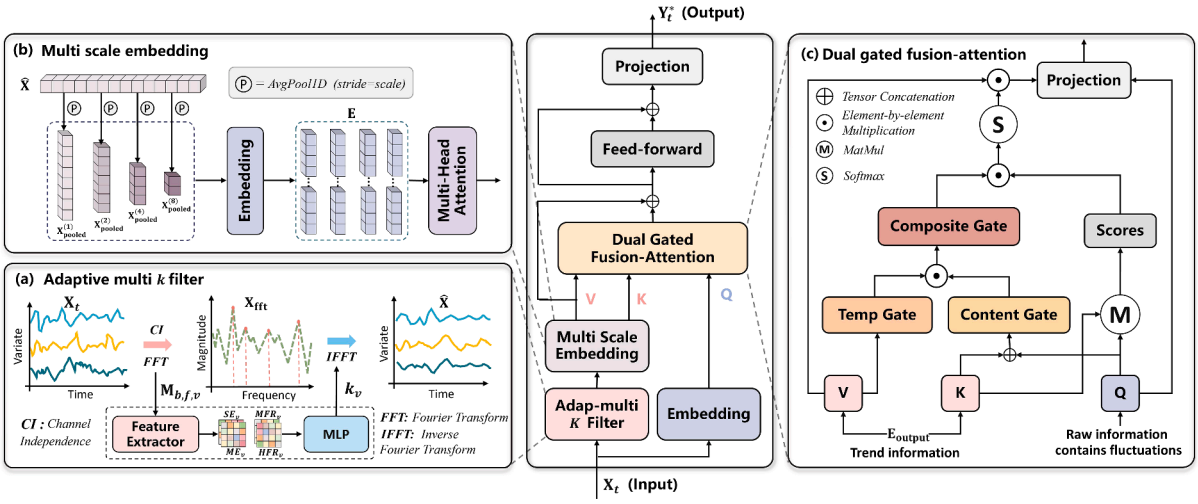


Fig. 1. The overall framework of AGformer. The input signal is first denoised via the adaptive multi-k filter (AKF in (a)) to extract clean temporal features. These features are then encoded using the multi-scale embedding (MSED in (b)) layer to capture multi-scale patterns. Finally, the MSED output is combined with the original data through the dual-gated fusion-attention (DGFA in (c)) mechanism, enabling the model to jointly model long-term dependencies and short-term fluctuations for accurate forecasting.

based on the attention score. Finally, encoders containing the Dual-Gated Fusion-Attention (DGFA) mechanism conduct collaborative forecasting of the filtered trend information and the raw information containing short-term fluctuations. Detailed implementations of each module are elaborated in the following subsections.

In this paper, we define three key information flows: (1) **Original data** ($\mathbf{X}_t$): the original input containing trends and short-term fluctuations; (2) **Filtered trend** ($\hat{\mathbf{X}}$): the denoised AKF output, retaining the main periodic components; (3) **Short-term fluctuations**: residual $\mathbf{X}_t - \hat{\mathbf{X}}$, which contains event-driven dynamics with standard denoising discard. In the DGFA module, the filtered trends are represented as K and V, while the raw data as Q explicitly reintroduces these fluctuations.

3.2. Adaptive multi_k filter

The AKF realizes discriminative filtering for multivariate time series via joint optimization in the frequency-domain, which preserves effective information while improving denoising performance. Fig. 1(a) illustrates details of AKF. The input time-domain signal $\mathbf{X}_t \in \mathbb{R}^{B\times T\times V}$ (where B denotes the batch size) is first transformed into the frequency domain via the Real-valued Fast Fourier Transform (RFFT) (Zhou et al., 2024) and the amplitude matrix ($M_{b,f,v}$) is obtained as follows:

$$\mathbf{X}_{\text{fft}} = \text{RFFT}(\mathbf{X}_t, \text{dim} = 1) \in \mathbb{C}^{B\times F\times V} \tag{2}$$

$$M_{b,f,v} = |\mathbf{X}_{\text{fft}}[b, f, v]| + \epsilon \tag{3}$$

where RFFT preserves only non-redundant frequency components, retaining $F = \lfloor T/2 \rfloor + 1$ frequency points for an input sequence of length T. b, f, v denote a specific value in B, F, V, respectively. A small constant $\epsilon > 0$ is added to $M_{b,f,v}$ to ensure numerical stability when computing the normalized probability distribution. Next, the TopK (Zhou et al., 2024) filtering strategy that preserves the largest k magnitudes in $M_{b,f,v}$ denoted as $\mathbf{Mask}[b, f, v]$, is applied to the frequency domain representation $\mathbf{X}_{\text{fft}}$, as shown below:

$$\mathbf{Mask}[b, f, v] = \begin{cases} 1, & \text{if } f \in \text{TopK}(M_{b,f,v}, K) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

$$\mathbf{X}_{\text{filtered}} = \mathbf{X}_{\text{fft}} \odot \mathbf{Mask}[b, f, v] \tag{5}$$

where $\odot$ is the element-by-element multiplication. Then the filtered frequency domain signal $\mathbf{X}_{\text{filtered}}$ is restored to the time domain signal $\hat{\mathbf{X}}$ through the Inverse Real-valued Fast Fourier Transform (IRFFT) (Zhou et al., 2024) as follows:

$$\hat{\mathbf{X}} = \text{IRFFT}(\mathbf{X}_{\text{filtered}}, \text{dim} = 1) \in \mathbb{R}^{B\times T\times V} \tag{6}$$

In Eq. (4), $K = \{k_v, v = 1, 2, \ldots, V\}$ is a list of V numbers, each corresponding to the number of frequency domain components that need to be preserved for each variable. In order to generate K more reasonably, this paper proposes four multivariate-adaptive frequency domain features (ME_v, SE_v, MFR_v, and HFR_v) suitable for time series forecasting. A simple multilayer perceptron (MLP (Goodfellow et al., 2016)) is used to obtain K based on the four features with the following formulas:

$$K = \text{MLP}\Big(\big\{ME_v, SE_v, MFR_v, HFR_v, v = 1, 2, \ldots, V\big\}\Big) \cdot (F - 1) \tag{7}$$

In practice, the MLP outputs continuous values in the range $(0, F-1)$. To obtain discrete frequency indices for filtering, we apply a rounding operation $\hat{K} = \text{round}(K)$ followed by clamping to the valid range $[1, F-1]$. The rounded $\hat{K}$ is then used as the number of top frequency components to retain in Eq. (4). This discrete sparsity level per variable is the key distinction of AKF from continuous spectral attenuation methods. First, the mean energy (ME_v) is proposed to reflect the average strength of the overall signal for the vth variable.

$$ME_v = \frac{1}{B \cdot F} \sum_{b=1}^{B} \sum_{f=1}^{F} \mathbf{M}_{b,f,v} \tag{8}$$

Signals with high ME_v manifest in the frequency domain as frequency components with large amplitudes. The corresponding periodic and repetitive patterns of different variables can be effectively learned by the MLP. Next, the spectral entropy (SE_v) is proposed to characterize the complexity of the spectral distribution for the vth variable as follows:

$$SE_v = -\frac{1}{B} \sum_{b=1}^{B} \sum_{f=1}^{F} p_{b,f,v} \log(p_{b,f,v}) \tag{9}$$

where the probability distribution $p_{b,f,v}$ of the energy is denoted by:

$$p_{b,f,v} = \frac{\mathbf{M}_{b,f,v}}{\sum_{i=1}^{F} \mathbf{M}_{b,i,v}} \tag{10}$$

Higher SE_v indicates a more complex signal with more widely dispersed energy, enabling the MLP to effectively learn the non-smoothness inherent in the data of the vth variable. Then the main frequency energy ratio

(MFR_v) is proposed to quantify the level of energy concentration, as defined by:

$$MFR_v = \frac{1}{B}\sum_{b=1}^{B}\frac{\sum_{k\in \text{top3}} \mathbf{M}_{b,f_k,v}}{\sum_{f=1}^{F}\mathbf{M}_{b,f,v}} \tag{11}$$

Higher MFR_v reflects that the energy of the vth variable is concentrated in the fundamental frequency and its harmonics. Consequently, the MLP can effectively capture the strong periodic component inherent in the data of the vth variable. Finally, the energy ratio in the latter 1/3 band is proposed to characterize the high frequency energy ratio (HFR_v), as demonstrated below:

$$HFR_v = \frac{1}{B}\sum_{b=1}^{B}\frac{\sum_{f=\lceil 2F/3\rceil}^{F}\mathbf{M}_{b,f,v}}{\sum_{f=1}^{F}\mathbf{M}_{b,f,v}} \tag{12}$$

Higher HFR_v means that the vth variable data contains rapidly changing information, enabling the MLP to effectively capture these rapid fluctuations. The above four frequency-domain features enable the MLP to learn the data properties of different variables.

3.3. Multi-scale embedding

To better capture the fine-grained changes in filtered trend information, we design a Multi-Scale Embedding (MSED) layer to realize the effective fusion of multiscale information. As illustrated in Fig. 1(b), the MSED comprises three key components:

1) **Multi-Scale Pooling**: This operation is achieved by AvgPool1D (Goodfellow et al., 2016), down-sampling filtered trend information at different temporal resolutions to extract multiple patterns. When the length of the time series (T) is not divisible by the defined down-sampling scale size (s), the tail elements would be truncated. Therefore, a Padding (Goodfellow et al., 2016) operation is first performed on the data $\hat{\mathbf{X}}$ before multiscale down-sampling to achieve feature dimension alignment across resolutions. For a given scale set $S = \{scale_1, scale_2, ..., scale_L\}$ (where L is the number of scales) corresponding to the multiple patterns being extracted, the down-sampling procedure for each scale $s \in S$ is formalized as:

$$p = (s - (T \bmod s)) \tag{13}$$

$$\mathbf{X}^{(s)} = \begin{cases} \text{Padding}(\hat{\mathbf{X}}, \text{pad_len} = p) & \text{if } s > 1 \\ \hat{\mathbf{X}} & \text{if } s = 1 \end{cases} \bmod s \tag{14}$$

$$\mathbf{X}^{(s)}_{\text{pooled}} = \text{AvgPool1D}\left(\mathbf{X}^{(s)}\right) \in \mathbb{R}^{B\times\lceil T/s\rceil\times V} \tag{15}$$

where $\lceil T/s\rceil$ is an upward rounding operation on T/s.

2) **Shared Embedding Layer**: The multi-scale features obtained above are projected into a unified high-dimensional space through a shared embedding network, defined as follows:

$$\mathbf{E}^{(s)} = \text{Linear}\left(\mathbf{X}^{(s)}_{\text{pooled}}\right) \in \mathbb{R}^{B\times V\times d_{\text{model}}} \tag{16}$$

$$\mathbf{E} = \text{Concat}\left(\mathbf{E}^{(s)}\right) \in \mathbb{R}^{B\times L\times V\times d_{\text{model}}} \tag{17}$$

where d_{model} is the embedding dimension, and $\mathbf{E}^{(s)}$ is the overall representation of the embedding vectors corresponding to multiple scales. $\mathbf{E}$ is the result of a simple concatenation of the embedding vectors in $\mathbf{E}^{(s)}$.

3) **Multi-Scale Attention Fusion Layer**: This module extends the conventional multi-head attention mechanism (Vaswani et al., 2017) by averaging the query (Q_E) on the scale dimension to generate a global representation.

$$Q_E \in \mathbb{R}^{B\times V\times d_{\text{model}}} = \frac{1}{L}\sum_{s=1}^{L}\mathbf{E} \tag{18}$$

Subsequently, inter-scale attention scores (score) are computed to dynamically fuse multi-scale features.

$$\text{score} = \frac{Q_E \cdot K_E}{\sqrt{d_{\text{head}}}} \tag{19}$$

where Q_E is the query, K_E is the key and d_{head} is the number of attention heads of the attention fusion layer. Compared to simplistic concatenation (Zhou et al., 2020) or weighted fusion strategies (Wu et al., 2021), this approach exhibits enhanced dynamic adaptability and broader inter-scale interactions, thereby strengthening cross-scale information integration. Finally, the output of this layer can be represented as E_{output}.

$$\text{E}_{\text{output}} = \text{Linear}\left(\text{Softmax}(\text{score}) \cdot V_E\right) \in \mathbb{R}^{B\times V\times d_{\text{model}}} \tag{20}$$

where V_E is the value of the attention fusion layer.

3.4. Dual-gated fusion-attention

While E_{output} retains essential filtered trend information, a Dual-Gated Fusion-Attention (DGFA) module is proposed to mitigate potential loss of localized detail fluctuations during filtering. This module utilizes E_{output} as K (key) and V (value), along with the raw data containing detailed fluctuations as Q (query). The DGFA then integrates some of the volatility information (regarded as "noise") into the filtered trend information, ensuring that global trends and localized detail fluctuations are fully retained. The dual gating comprises temporal gating and content gating. The temporal gating (TempGate) applies grouped convolution to V, independently processing the temporal dimension for each attention head, as shown below:

$$TG_{\text{output}} = \text{TempGate}(V) = \sigma(\text{DSConv1D}(V)) \tag{21}$$

where, σ is the Sigmoid function, mapping the convolutional output to the $[0, 1]$ interval to represent *importance weights* for each temporal position. DSConv1D is the 1D depth-separable convolutional (Goodfellow et al., 2016). This operation captures local temporal patterns and enhances sensitivity to critical time windows. The content gating (ContentGate) concatenates Q and K, which are then projected onto a scalar through a linear layer and similarly activated by a Sigmoid function, as shown below:

$$CG_{\text{output}} = \text{ContentGate}(Q, K) = \sigma(\text{Linear}[Q \oplus K]) \tag{22}$$

where $\oplus$ is the tensor concatenation operation. This mechanism dynamically evaluates the inter-feature relevance between the two input features. A composite gating (CompositeGate) mechanism is designed to fuse TG_{output} and CG_{output}, simultaneously capturing temporal patterns and highlighting cross-timestep semantic dependencies through adaptive weighting as follows:

$$DG_{\text{output}} = \text{CompositeGate}(TG_{\text{output}}, CG_{\text{output}}) = TG_{\text{output}} \odot CG_{\text{output}}, \tag{23}$$

where $\odot$ denotes element-wise multiplication. This multiplicative interaction simultaneously enforces temporal locality (via TempGate) and semantic relevance (via ContentGate), acting as a spatiotemporal validity check before attention weighting. The final attention weights ($Attns$) are derived through element-wise multiplication of the original attention scores and gating weights, enabling feature selection as follows:

$$Attns = \text{Dropout}(\text{Softmax}\left(\frac{QK^{\top}}{\sqrt{d}} \odot DG_{\text{output}}\right)) \tag{24}$$

where d is the number of attention heads. The Dropout process follows Softmax to regularize $Attns$. Finally, the output of the module ($DGFA_{\text{output}}$) can be obtained from $Attns$ as follows:

$$DGFA_{\text{output}} = \text{Proj}((Attns \odot V) \oplus Q) \tag{25}$$

where Proj is the projection operation.

Theoretical Rationale. DGFA's dual-gated pathways complement each other: TempGate detects local temporal windows, such as the onset of abrupt structural breaks via deep separable convolutions, while ContentGate evaluates semantic relevance (position-invariant) based on Q and K. The two are multiplied at the element level via CompositeGate

Table 1
Basic information of datasets.

Dataset	Dim	Frequency	Dataset Size	Domain
ETTh1	7	Hourly	(8545, 2881, 2881)	Electricity
ETTh2	7	Hourly	(8545, 2881, 2881)	Electricity
ETTm1	7	15min	(34465, 11521, 11521)	Electricity
ETTm2	7	15min	(34465, 11521, 11521)	Electricity
Electricity	321	Hourly	(26304, 2633, 5261)	Electricity
Exchange	8	Daily	(5120, 665, 1422)	Economy
Weather	21	10min	(36792, 5271, 10540)	Weather
PEMS04	307	5min	(10172,3375,3375)	Transportation
PEMS08	170	5min	(10690,3548,3548)	Transportation

to achieve spatiotemporal joint gating. This mechanism is analogous to the complementary gates in LSTM (Hochreiter & Schmidhuber, 1997), but operates within a cross-attention framework, thereby preserving both global trends and local fluctuations simultaneously.

4. Experiments

4.1. Experiments setting

Datasets: This study employs multiple publicly available real-world time-series benchmark datasets for forecasting. These datasets span multiple domains with varying temporal characteristics: strong periodicity (ETT, PEMS), weak periodicity (Exchange), and mixed dynamics (Weather). They collectively cover a wide range of variable counts (8 to 321) and sampling frequencies (5-min to daily), providing a comprehensive testbed for evaluating generalization across diverse forecasting scenarios. Detailed information for each dataset is provided in Table 1 (Liu et al., 2023), where Dim denotes the number of variables, and $DatasetSize$ indicates the total number of time points in the training, validation, and test sets, respectively. The specific details are as follows:

- Electricity: ETT series dataset (Zhou et al., 2020) originates from China's power transformer monitoring system, focusing on oil temperature (OT) and active/reactive power data from different sides of transformers (high, medium, low). Its hourly-level (ETTh1/ETTh2) and 15-min-level (ETTm1/ETTm2) subsets reflect equipment operational status at varying temporal resolutions. The specific geographic locations of the sensors are not publicly disclosed. Electricity dataset (Trindade, 2015) records hourly load data for 370 users across Europe and North America. Each user unit represents a variable, revealing complex electricity consumption patterns between residential and commercial entities from 2011 to 2014.
- Economy: Exchange rate dataset contains daily exchange rates for eight currencies against the US dollar, covering major global economies and exhibiting distinct financial time series characteristics.
- Weather: Weather dataset, collected from wind farm regions in the United States, includes 21 meteorological variables (such as barometric pressure, temperature, and humidity), reflecting climatic fluctuations in mid-latitude regions.
- Transportation: PEMS04 and PEMS08 represent California traffic flow data, comprising 307 and 170 monitoring stations respectively. These datasets record vehicle speed, flow rate, and occupancy at 5-min intervals, revealing complex spatio-temporal dependency patterns across urban road networks during peak hours and other time periods.

Baselines and Selection Justification : To ensure a comprehensive and fair comparison, we compare AGformer with eleven state-of-the-art models from three dominant architectural paradigms. The selection of these baselines is justified as they are widely recognized and have demonstrated strong performance in recent multivariate time series forecasting literature, covering the key methodological trends:

Table 2
Evalution metrics.

Metric	Formula	Sensitive to Large Errors	Parameter Explanation
MAE	$\frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$	×	n: number of samples y_i: true value $\hat{y}_i$: predicted value
MSE	$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	✓	

Table 3
Hyperparameter settings for the AGformer.

Datasets	seq_len	batch_size	d_model	n_scale	window_size
ETTh1	512	64	128	4	(8,10,12)
ETTh2	512	16	128	4	(8,10,12)
ETTm1	336	16	256	4	(8,10,12)
ETTm2	512	16	512	4	(8,10,12)
Electricity	336	16	512	4	4
Exchange	96	32	256	4	(4,10)
Weather	512	64	512	4	(4,8)
PEMS04	96	8	32	4	4
PEMS08	336	8	64	4	4

- Transformer-based Models: We include iTransformer (Liu et al., 2023) for its inverted structure that treats variates as tokens; FEDformer (Zhou et al., 2022) as a representative of frequency-enhanced Transformers; Crossformer (Zhang & Yan, 2023) for its cross-dimension dependency modeling; Stationary (Liu et al., 2022) for its explicit handling of non-stationarity; and TQNet (Lin et al., 2025) for its learnable vectors with periodic shifts. These models represent the evolution of attention-based architectures.
- Mamba-based Models: We include SST (Xu et al., 2025) for its hybrid Mamba-Transformer expert architecture with linear complexity $O(L)$ and S-Mamba (Wang et al., 2024b) as a representative Mamba-based forecaster. These models represent the emerging SSM paradigm for time series forecasting.
- TCN-based Models: We select TimesNet (Wu et al., 2022) for its powerful 2D-variation modeling based on converting time series into a set of 2D tensors, which is a leading model in capturing multi-periodicity.
- MLP-based Models: We include DLinear (Zeng et al., 2023) for its simplicity and strong performance that often rivals complex Transformers, and TimeMixer (Wang et al., 2024a), as a recent advanced model that employs decomposable multiscale mixing. These models represent the modern, efficient linear paradigm.

Experimen tal results reflect the best performance selected from prediction horizons of {96, 192, 336, 720} to ensure a fair comparison.

Evaluation Metrics: We employ Mean Absolute Error (MAE) and Mean Squared Error (MSE) as evaluation metrics for forecasting tasks. MAE measures the average of absolute errors between predicted and true values, reflecting the mean deviation of predictions from actual observations. In contrast, MSE calculates the average of squared errors, emphasizing larger prediction deviations due to its quadratic penalty. Table 2 summarizes the mathematical formulations, parameter definitions, and characteristics of these metrics.

Implementation Details: All experiments are conducted using the PyTorch framework on a single NVIDIA RTX 4090D 24 GB GPU, with models trained using the classic Adam optimizer.

Hyperparameter Analysis: The hyperparameter settings for the AGformer model across different datasets are shown in Table 3. The results for mainstream models used in this paper are sourced from https://decisionintelligence.github.io/OpenTS/result/#multivariate_forecasting. The seq_len for each model is selected from the optimal results among (96, 336, 512), and this paper follows the same convention. The selection of $window_size$ for the multi-scale

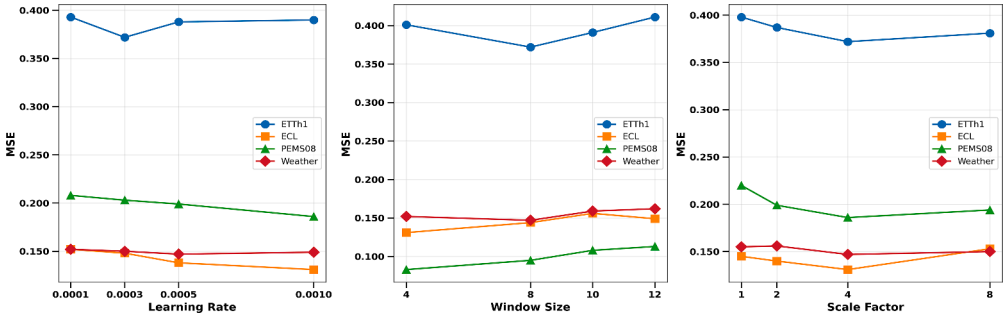


Fig. 2. Sensitivity of hyperparameters to learning rate, filter size, and number of scales. Results recorded under conditions of historical window length seq_len = 96 and prediction window length pred_len = 96.

window is based on the divisibility relationship between the sampling frequency and the periodic structure of each dataset, ensuring that downsampled sequences remain aligned with natural rhythms. For the hourly ETTh1/ETTh2 datasets (24-step daily cycle) and the 15-min ETTm1/ETTm2 datasets (96-step daily cycle), the window sizes are set to $\{8, 10, 12\}$; here, 8 and 12 are exact divisors, while 10 approximates a standard workday duration, enabling the model to capture daily periodic patterns across different resolutions. For the high-dimensional ECL and PEMS datasets, which exhibit weaker periodicity, smaller window sizes such as $\{4\}$ or $\{4, 10\}$ are employed to prevent excessive sequence compression and noise amplification. For the Weather dataset (10-min sampling, 144-step daily cycle), the sizes $\{4, 8\}$ are chosen; both values are divisors of 144, thereby effectively embedding the daily periodic structure. The final values are tuned via grid search on the validation set for each dataset. The DGFA's Dual-Gate mechanism is activated via a Sigmoid function, resulting in a gating threshold range of $[0, 1]$ (see Fig. 2).

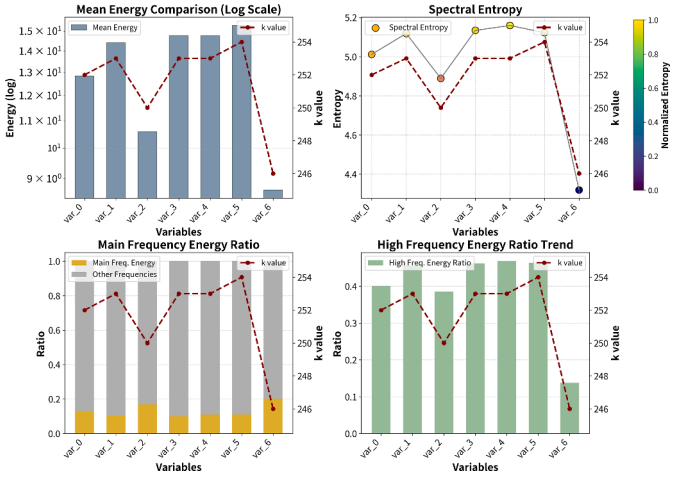


Fig. 3. Frequency-domain features and multivariate k-values in the ETTh2 dataset.

4.2. Experiments results

Table 4 summarizes the forecasting performance of AGformer and the benchmark models. As evidenced by the results, the proposed AGformer achieves performance improvements across all datasets, securing 29 and 26 best results (out of 44 total metrics) in both MSE and MAE evaluations—demonstrating superior generalization capabilities over existing approaches. On the ETTm2 dataset, AGformer's MSE and MAE are reduced by an average of 8.1% to 23.6% compared to other SOTA models, including an average of 15.1% and 15.8%. In addition, a comparison with DLinear—which obtains 6 best results in MSE and 9 in MAE—further highlights AGformer's stronger ability to maintain both prediction accuracy and robustness across diverse temporal patterns. Beyond these established baselines, we further compare AGformer against three recent 2025 architectures: TQNet, SST, and S-Mamba.Results for these models are sourced from their original publications or reproduced under identical settings. AGformer consistently outperforms these latest baselines across all datasets: compared to TQNet, SST, and S-Mamba, AGformer achieves average MSE reductions of 2.4%–17.9%, 1.8%–23.2%, and 2.3%–19.2% across the nine benchmarks, respectively. These results demonstrate AGformer's competitiveness against even the latest specialized models, validating that its adaptive filtering and gated fusion-attention design remains effective amidst the rapid evolution of forecasting architectures. On the ETT series datasets, AGformer achieves an average reduction of 9.2% in MSE and 5.3% in MAE. On the Weather dataset, it further reduces MSE and MAE by 7.9% and 10.9% on average, respectively.

For traffic flow datasets, AGformer exhibits strong performance, particularly on PEMS08. Notably, on PEMS08, AGformer achieves the best MSE performance across all prediction horizons (96, 192, 336, 720), with an average MSE of 0.254, outperforming all benchmark models including TQNet, SST, and S-Mamba. On PEMS04, however, while TimesNet achieves the lowest MSE and MAE values, AGformer remains competitive, surpassing Crossformer by 20.4%, TimeMixer by 37.1%, DLinear by 35.8%, and Stationary by 41.8% in MSE. This performance discrepancy can be attributed to the distinct traffic characteristics of the two regions. PEMS04 covers more urbanized areas with higher rates of non-recurring events such as accidents and weather disruptions, where TimesNet's 2D transformation effectively captures such transient variations. In contrast, PEMS08 exhibits stronger periodicity and lower noise levels, where AKF's adaptive filtering and MSED's multi-scale fusion yield superior performance by preserving dominant periodic components while suppressing irrelevant noise. These results highlight AGformer's effectiveness in capturing complex spatiotemporal dependencies in periodic traffic data, while also suggesting that future work could enhance its robustness to highly irregular data by integrating event-detection mechanisms.

Finally, compared to Transformer-based models, AGformer achieves an average reduction of 4.9% to 17.9% in both MSE and MAE. The consistent performance improvements of AGformer across electricity, finance, weather, and transportation domains underscore its potential universality for multivariate long-term forecasting. In summary, AGformer's multivariate adaptive strategy and enhanced capability for long- and short-term information interaction make it well-suited for these tasks, with demonstrated effectiveness extending across diverse application domains.

AGformer achieves state-of-the-art performance among specialized small-scale models, it is not appropriate to directly compare its MSE/MAE metrics with those of time-series foundation models like Sundial and Chronos-2; the former undergoes supervised training from scratch on each dataset, whereas the latter relies on zero-shot inference

Table 4
Comparative analysis of proposed model and benchmark models.

Models		AGformer		iTransformer		FEDformer		Crossformer		Stationary		TQNet		SST		S-Mamba		TimesNet		DLinear		TimeMixer	
Datasets	len	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.372**	0.401	0.386	0.405	0.379	0.419	0.411	0.435	0.591	0.524	**0.372**	0.393	0.381	0.405	0.392	**0.390**	0.389	0.412	0.379	0.403	**0.372**	0.401
	192	**0.406**	0.422	0.424	0.440	0.420	0.444	0.409	0.438	0.615	0.540	0.428	0.426	0.430	0.434	0.449	0.439	0.440	0.443	0.408	**0.419**	0.413	0.430
	336	**0.428**	**0.440**	0.449	0.460	0.458	0.466	0.433	0.457	0.632	0.551	0.476	0.446	0.443	0.446	0.467	0.481	0.523	0.487	0.440	**0.440**	0.438	0.450
	720	**0.455**	0.476	0.495	0.487	0.474	0.488	0.501	0.514	0.828	0.658	0.487	0.470	0.502	0.501	0.475	**0.468**	0.521	0.495	0.471	0.493	0.486	0.484
	Avg	**0.415**	**0.434**	0.439	0.448	0.433	0.454	0.439	0.461	0.667	0.568	0.441	**0.434**	0.439	0.447	0.446	0.445	0.468	0.459	0.425	0.439	0.427	0.441
ETTh2	96	**0.279**	**0.343**	0.297	0.348	0.337	0.380	0.728	0.603	0.347	0.387	0.295	**0.343**	0.291	0.346	0.292	0.357	0.334	0.370	0.300	0.364	0.281	0.351
	192	**0.347**	**0.386**	0.372	0.403	0.415	0.428	0.723	0.607	0.379	0.418	0.367	0.393	0.369	0.397	0.380	0.402	0.404	0.413	0.387	0.423	0.349	0.387
	336	0.373	**0.412**	0.388	0.417	0.389	0.457	0.740	0.628	**0.358**	0.413	0.417	0.427	0.374	0.414	0.391	0.420	0.389	0.435	0.490	0.487	0.366	0.413
	720	0.403	0.439	0.424	0.444	0.483	0.488	1.386	0.882	0.422	0.457	0.433	0.446	0.419	0.447	0.437	0.455	0.434	0.448	0.704	0.597	**0.401**	**0.436**
	Avg	0.351	**0.395**	0.370	0.403	0.406	0.438	0.894	0.680	0.377	0.419	0.378	0.402	0.363	0.401	0.375	0.409	0.390	0.416	0.470	0.468	**0.349**	0.397
ETTm1	96	**0.288**	**0.342**	0.300	0.353	0.463	0.463	0.314	0.367	0.415	0.410	0.311	0.353	0.298	0.355	0.311	0.380	0.340	0.378	0.300	0.345	0.293	0.345
	192	**0.328**	**0.365**	0.341	0.380	0.575	0.516	0.374	0.410	0.494	0.451	0.356	0.378	0.347	0.381	0.389	0.419	0.392	0.404	0.336	0.366	0.335	0.372
	336	**0.362**	**0.386**	0.374	0.396	0.618	0.544	0.413	0.432	0.577	0.490	0.390	0.401	0.374	0.397	0.401	0.417	0.423	0.426	0.367	**0.386**	0.368	**0.386**
	720	**0.419**	0.422	0.429	0.430	0.612	0.551	0.753	0.613	0.636	0.535	0.452	0.440	0.429	0.428	0.488	0.476	0.475	0.453	**0.419**	**0.416**	0.426	0.417
	Avg	**0.349**	0.379	0.361	0.390	0.567	0.519	0.464	0.455	0.531	0.471	0.377	0.393	0.362	0.390	0.397	0.423	0.408	0.415	0.356	**0.378**	0.356	0.380
ETTm2	96	0.165	**0.255**	0.175	0.266	0.216	0.309	0.296	0.391	0.210	0.294	0.173	0.256	0.176	0.264	0.191	0.301	0.189	0.265	**0.164**	**0.255**	0.165	0.256
	192	**0.221**	**0.294**	0.242	0.312	0.297	0.360	0.369	0.416	0.338	0.373	0.238	0.298	0.231	0.303	0.253	0.312	0.254	0.310	0.224	0.304	0.225	0.298
	336	**0.276**	**0.331**	0.282	0.337	0.366	0.400	0.588	0.600	0.432	0.416	0.301	0.340	0.290	0.339	0.298	0.342	0.313	0.345	0.277	0.337	0.277	0.332
	720	**0.358**	**0.382**	0.375	0.394	0.459	0.450	0.750	0.612	0.554	0.476	0.397	0.396	0.388	0.398	0.409	0.407	0.413	0.351	0.371	0.401	0.360	0.387
	Avg	**0.255**	**0.316**	0.268	0.327	0.335	0.380	0.501	0.505	0.384	0.390	0.277	0.323	0.271	0.326	0.288	0.341	0.331	0.345	0.259	0.324	0.257	0.318
ECL	96	**0.131**	**0.225**	0.134	0.230	0.191	0.305	0.134	0.231	0.171	0.274	0.134	0.229	0.141	0.239	0.157	0.255	0.169	0.271	0.140	0.237	0.153	0.256
	192	0.148	**0.241**	0.154	0.250	0.203	0.316	**0.146**	0.243	0.180	0.283	0.154	0.247	0.159	0.255	0.188	0.271	0.180	0.280	0.154	0.251	0.168	0.269
	336	**0.163**	**0.258**	0.169	0.265	0.221	0.333	0.165	0.264	0.204	0.305	0.169	0.264	0.171	0.268	0.192	0.275	0.204	0.304	0.169	0.268	0.189	0.291
	720	0.201	0.290	**0.194**	**0.288**	0.259	0.364	0.237	0.314	0.221	0.319	0.201	0.294	0.208	0.300	0.241	0.339	0.205	0.304	0.204	0.301	0.228	0.320
	Avg	**0.160**	**0.253**	0.163	0.258	0.219	0.330	0.170	0.263	0.194	0.295	0.164	0.259	0.170	0.266	0.195	0.285	0.189	0.290	0.167	0.264	0.185	0.284
Exchange	96	0.083	**0.202**	0.086	0.205	0.138	0.268	0.231	0.356	0.121	0.247	0.083	**0.202**	0.086	0.205	0.087	0.207	0.112	0.242	**0.080**	**0.202**	0.084	0.207
	192	0.176	0.299	0.177	0.299	0.273	0.379	0.460	0.509	0.220	0.337	**0.175**	**0.296**	0.179	0.301	0.180	0.302	0.209	0.334	0.182	0.321	0.178	0.300
	336	**0.327**	**0.414**	0.331	0.417	0.437	0.485	1.034	0.825	0.352	0.437	0.328	0.415	0.331	0.419	0.330	0.416	0.358	0.435	**0.327**	0.434	0.376	0.451
	720	0.836	0.688	0.846	0.693	1.158	0.828	1.576	1.021	0.725	0.656	0.840	0.691	0.885	0.713	0.856	0.698	0.944	0.736	**0.578**	**0.605**	0.884	0.707
	Avg	0.355	0.401	0.360	0.403	0.501	0.490	0.825	0.678	0.355	0.419	0.357	0.401	0.370	0.410	0.363	0.406	0.406	0.437	**0.292**	**0.391**	0.381	0.416
Weather	96	0.147	**0.196**	0.157	0.207	0.229	0.298	**0.143**	0.210	0.188	0.242	0.157	0.200	0.153	0.205	0.169	0.221	0.168	0.214	0.170	0.230	0.147	0.198
	192	**0.190**	**0.239**	0.200	0.248	0.265	0.334	0.195	0.261	0.240	0.290	0.206	0.245	0.196	0.244	0.205	0.248	0.219	0.262	0.216	0.273	0.192	0.243
	336	**0.243**	**0.280**	0.252	0.287	0.330	0.372	0.254	0.319	0.322	0.328	0.262	0.287	0.246	0.283	0.288	0.299	0.278	0.302	0.258	0.307	0.247	0.284
	720	**0.311**	**0.329**	0.320	0.336	0.423	0.418	0.335	0.385	0.396	0.378	0.344	0.342	0.314	0.334	0.335	0.369	0.353	0.351	0.323	0.362	0.318	0.330
	Avg	**0.223**	**0.261**	0.232	0.270	0.312	0.355	0.232	0.294	0.287	0.310	0.242	0.269	0.227	0.267	0.249	0.284	0.254	0.282	0.242	0.293	0.226	0.264
PEMS04	96	0.119	0.231	0.125	0.230	0.598	0.571	0.122	0.236	0.192	0.304	0.133	0.238	0.153	0.244	0.131	0.249	**0.113**	**0.214**	0.197	0.295	0.198	0.299
	192	0.149	0.255	0.145	0.249	0.996	0.804	0.173	0.271	0.232	0.333	0.155	0.262	0.172	0.264	0.186	0.266	**0.128**	**0.230**	0.214	0.310	0.216	0.312
	336	0.153	0.261	0.158	0.263	1.817	1.067	0.208	0.307	0.245	0.332	0.165	0.281	0.221	0.304	0.206	0.283	**0.132**	**0.235**	0.237	0.327	0.257	0.344
	720	0.202	0.309	0.195	0.300	1.047	0.795	0.279	0.362	0.401	0.432	0.225	0.333	0.266	0.345	0.249	0.331	**0.152**	**0.256**	0.324	0.392	0.321	0.395
	Avg	0.156	0.264	0.156	0.261	1.118	0.809	0.196	0.294	0.268	0.345	0.170	0.279	0.203	0.289	0.193	0.282	**0.131**	**0.234**	0.243	0.331	0.248	0.338
PEMS08	96	**0.186**	0.244	0.194	**0.235**	0.722	0.795	0.230	0.260	0.401	0.369	0.208	0.266	0.194	0.258	0.209	0.269	0.282	0.237	0.320	0.329	0.200	0.237
	192	**0.236**	0.275	0.288	**0.255**	1.159	0.585	0.239	0.264	0.353	0.420	0.239	0.288	0.242	0.287	0.246	0.286	0.344	0.264	0.379	0.353	0.273	0.260
	336	**0.256**	0.293	0.345	0.276	2.222	0.810	0.272	0.289	0.508	0.423	0.271	0.311	0.267	0.304	0.270	0.318	0.381	0.277	0.415	0.374	0.341	**0.264**
	720	**0.315**	0.347	0.366	0.318	1.243	1.142	0.320	0.316	0.568	0.543	0.329	0.362	0.331	0.359	0.327	0.358	0.375	**0.308**	0.483	0.437	0.356	0.325
	Avg	**0.254**	0.290	0.298	**0.271**	1.336	0.833	0.265	0.282	0.458	0.439	0.262	0.307	0.259	0.302	0.263	0.308	0.346	0.272	0.399	0.373	0.293	0.272
Optimal Count		29	25	1	4	0	0	2	0	1	0	2	4	0	0	0	2	5	6	6	9	3	3

The best results are highlighted in bold and the second best results are in underlined.

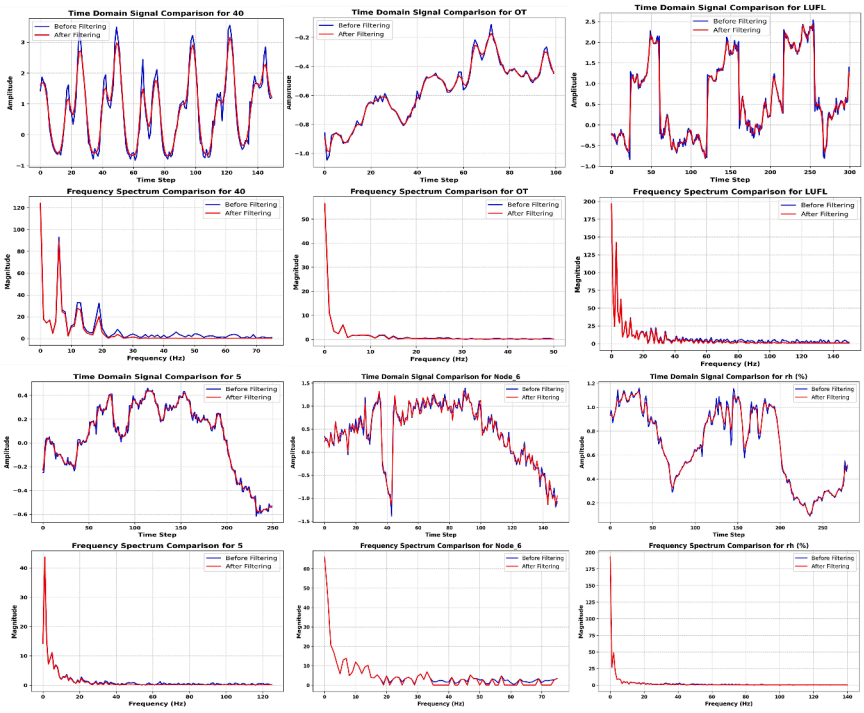


Fig. 4. Adaptive filtering time-frequency domain results.

following large-scale, cross-domain pre-training. TSFMs achieve cross-domain generalization with minimal adaptation costs, while AGformer is lightweight (0.42M parameters), supports domain adaptation, and incorporates explicit inductive biases to handle heterogeneous variables and sudden events—making it well-suited for domain-specific scenarios that prioritize granular control and low-latency deployment.

4.3. Model analysis

(1) Effectiveness Analysis of the AKF

This section analyzes the effectiveness of the AKF module, focusing on: 1) The intrinsic relationship between the adaptively generated multivariate k values and the frequency-domain properties of each variable. 2) The filtering performance of the AKF adaptive filter. Fig. 3 visually presents the frequency-domain characteristics of all variables in the ETTh2 dataset and their corresponding k-values. The four subplots correspond to the ME_v, SE_v, MFR_v, and HFR_v values for all variables, respectively. The red broken line represents the adaptively generated k value. For each dataset, the k-value list undergoes iterative optimization during training and is subsequently preserved for direct application during testing.

The blue bar chart displays the ME_v values for each variable, representing the average energy of each variable. The higher ME_v values for **Var3** and **Var5** indicate that these variables exhibit greater overall signal strength in the frequency domain, corresponding to channels with frequent fluctuations and significant amplitude variations. For such high-energy, highly fluctuating variables, the AKF module generates larger k values to specifically preserve their rich frequency components.

Scatter plots and trend charts display the SE_v values for each variable, where lighter shades indicate higher entropy values, reflecting more complex energy distributions across frequency bands. The high entropy values for **Var3** and **Var4** suggest that the signal energy for these variables is distributed uniformly and with high complexity across the frequency domain. This indicates more intricate and harder-to-predict temporal patterns, potentially associated with non-stationary channel dynamics. For such complex and highly unpredictable variables, the algorithm assigns larger k values to prudently preserve their frequency-domain complexity, avoiding loss of critical patterns due to excessive filtering.

The orange bar chart displays the MFR_v values for different variables. A lower value indicates that the main frequency component differs little from the others, corresponding to a certain periodicity. **Var2** and **Var6** exhibit higher MFR_v values, indicating pronounced periodicity. The algorithm generates smaller k values to perform stronger filtering, thereby highlighting their dominant periodic components.

The green histogram displays the HFR_v values for different variables, all falling between 0.4 and 0.5, indicating a moderate proportion of high-frequency energy. This reveals the presence of rapid fluctuations or abrupt jumps within the signal. Consequently, for variables exhibiting significant rapid changes (higher HFR_v), the AKF module assigns correspondingly larger k values to preserve these high-frequency details. These observations rigorously corroborate the analysis presented in Section 3.2.

Additionally, we selected different variables from six datasets to visualize the adaptive filtering performance of the AKF, as shown in Fig. 4. It is evident that AKF possesses the capability for adaptive filtering across diverse datasets. For instance, var40 from the Electricity dataset exhibits distinct periodicity, while Node6 from the PEMS dataset demonstrates significant noise and abrupt changes influenced by external disturbances and non-stationary inputs. Moreover, AKF achieves high-fidelity filtering across multiple classes of non-stationary time-series data without introducing noticeable lag or distortion, demonstrating its excellent adaptability and robustness across diverse application scenarios. In summary, the proposed AKF module can effectively capture both the frequency-domain features of different variables in the dataset and the temporal patterns they present in order to generate targeted filtering strategies.

(2) Effectiveness Analysis of the MSED

This section examines the effectiveness of the multi-scale embedding layer by plotting 1) UMAP maps, 2) multi-scale signals, 3) the distribution of embedded vectors, and 4) inter-scale attention weights.

Fig. 5 plots the UMAP mapping of the MSED for the Electricity dataset, showing the data distribution at different scales, with different colors representing different scales (Scale 1, Scale 2, Scale 4, Scale

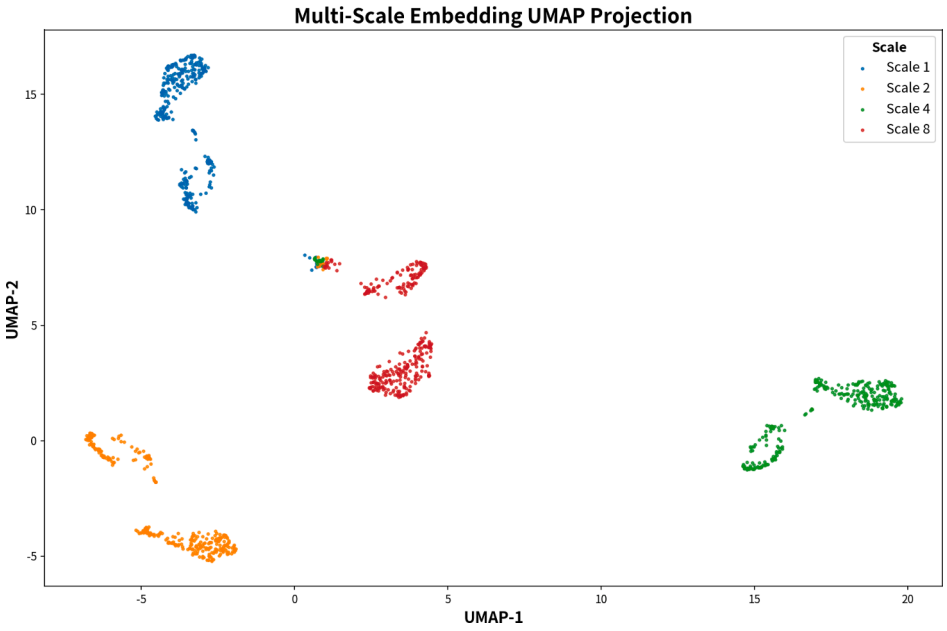


Fig. 5. Multi-scale embedding UMAP projection.

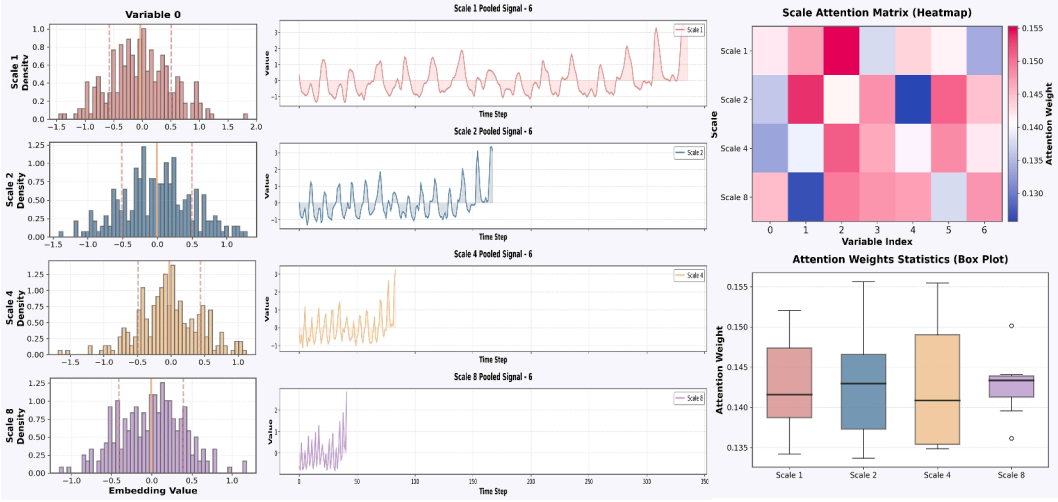


Fig. 6. Visualization of internal mechanisms via multi-scale embedding.

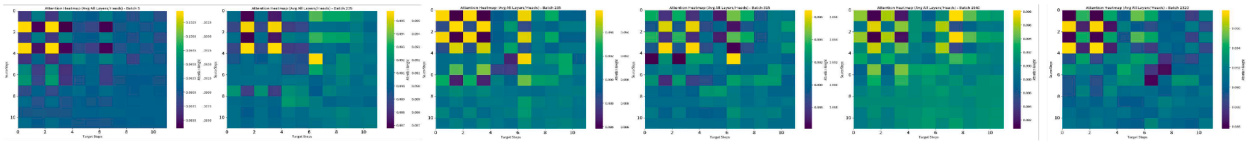
8). The visualization demonstrates that data points from each scale form relatively distinct clusters, indicating divergent feature patterns across temporal resolutions. This divergence underscores the MSED layer's capability to capture scale-specific temporal dynamics.

Furthermore, the inter-scale cluster separation is notably pronounced in Fig. 5. For instance, blue clusters (Scale 1) exhibit pronounced spatial concentration in the upper-left region, suggesting elevated volatility in this scale—a pattern consistent with short-term, high-frequency temporal features. From left to right, the data point distribution exhibits a hierarchical spatial structure: smaller scales (Scale 1, Scale 2) cluster predominantly on the left, while larger scales (Scale 4, Scale 8) spread toward the right. This spatial gradation reflects a progressive evolution from short-term, high-frequency patterns to long-term, low-frequency temporal characteristics across scales. Within each scale, clusters are partitioned into two compact sub-clusters. This recurrent clustering pattern suggests periodic variations persist across multiple temporal scales in the dataset, implying synchronized oscillatory behaviors embedded within both short- and long-term temporal resolutions.

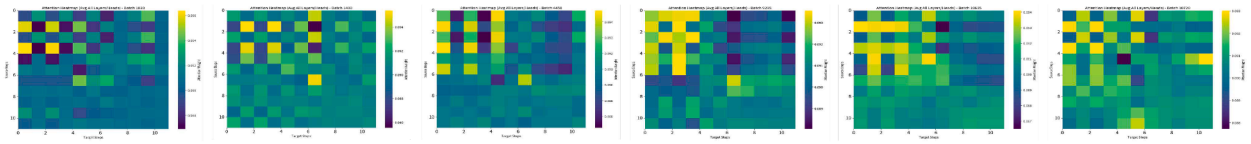
Fig. 6 illustrates the distribution of MSED's embedding vectors, the multiscale signals, and the distribution of inter-scale attention weights. Observing the embedding vector distribution plots reveals that the distributions across the four scales exhibit differences, yet the overall structure aligns with the data characteristics of the same dataset. Concurrently, the pooling signal plots demonstrate varying signal smoothness and trends extracted at different scales. This confirms that MSED effectively captures multi-temporal resolution features—including fine-grained short-period oscillations, medium-term cyclic patterns, and long-term trend shifts—without compromising the dataset's overall feature distribution. Additionally, it is evident that higher attention weights are assigned to smaller scales, indicating that the model prioritizes fine-grained temporal features—such as short-period oscillations and high-frequency transient patterns.

(3) Effectiveness Analysis of the DGFA

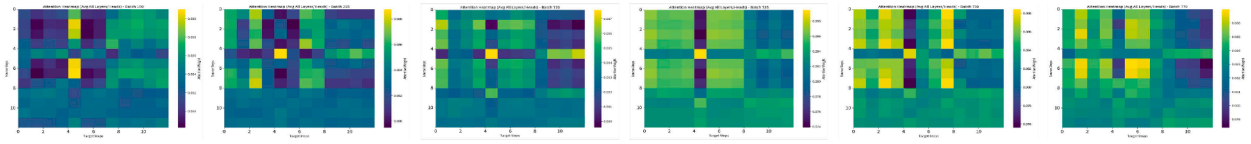
This section primarily analyzes the effectiveness of DGFA by plotting 1) attention maps of DGFA across different datasets, and 2) weight distribution diagrams and comparison diagrams of the two gates.



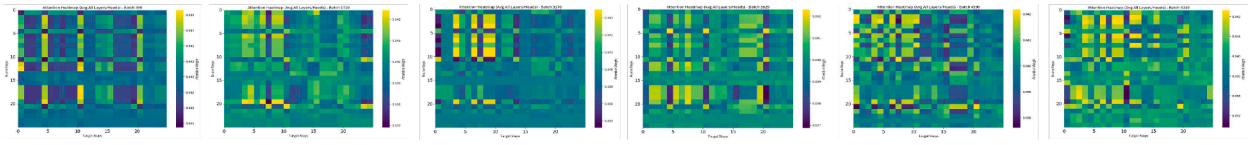
(a) Attention map for ETTh1 dataset



(b) Attention map for ETTm1 dataset



(c) Attention map for exchange dataset



(d) Attention map for weather dataset

Fig. 7. Attention maps for four datasets, each with six different samples from the test set.

Fig. 7 shows the attention maps of the four datasets ETTh1, ETTm1, Exchange, and Weather on the test set, respectively, which are plotted every number of batch, and six plotted results are randomly selected from all the plotted results. It can be seen that for each dataset, DGFA can accurately focus on the key information (the lighter the color, the more important the information is), and more yellow areas indicate that more and more important information can be captured.The patterns of ETTh1 and ETTm1 in the attention maps are very similar, which suggests that DGFA is able to capture the similarity between two datasets from the same data source but with different temporal granularity. The Weather dataset is a long time series dataset, and its attention map shows that DGFA can learn the periodic features of this dataset very well, suggesting that DGFA is also very suitable for long time series.

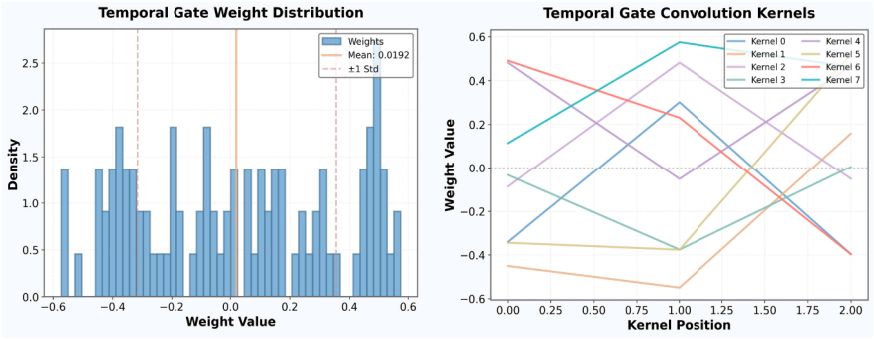
We analyzed the two gating branches of DGFA using the Electricity dataset. As shown in Fig. 8, TempGate generates a diverse set of 1D convolutional kernels corresponding to rising, falling, and symmetric patterns. Its weight distribution exhibits a relatively high mean (0.0192) and a large standard deviation (0.3352), indicating that it is highly sensitive to local temporal structures. In contrast, ContentGate concentrates the vast majority of its positive weights across a small number of dimensions, with the top 10 positive weights exhibiting high averages, while the remaining weights hover near zero. The mean of the overall weight distribution is only 0.0068, and the standard deviation is merely 0.0811. This complementary division of labor—where TempGate captures dynamic local patterns while ContentGate performs sparse feature selection—explains why DGFA can selectively extract those short-term fluctuation signals. Although these signals contain rich information, they are often suppressed in conventional denoising methods. A quantitative comparison of these two gating mechanisms further confirms the unique roles they each play.

4.4. Ablation experiments

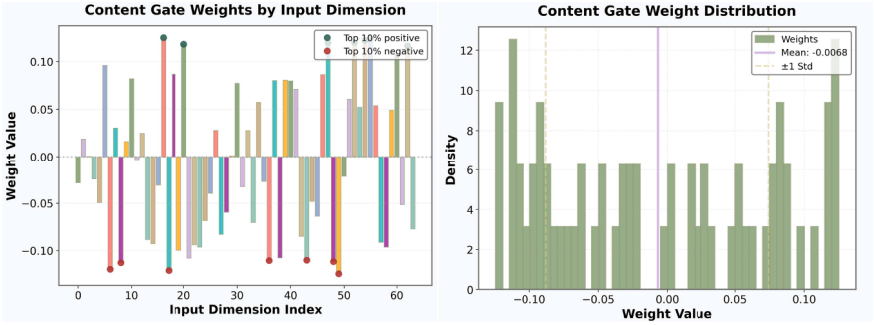
To thoroughly evaluate the contributions of the three proposed modules (AKF, MSED, DGFA), this section analyzes experimental results based on both individual and combined ablation tests. Specific data can be found in Tables 5 and 6. The experimental design encompasses three categories: single-module removal or replacement, combined-module removal (i.e., “retaining only X”), and DGFA internal mechanism ablation. All analyses are based on average MSE to ensure comprehensive and robust conclusions.

Firstly, the AKF module serves as a crucial filtering component, providing the model with flexible adaptive data processing capabilities. Direct removal of the AKF module resulted in a 3.19% decline in average performance. This component also contributes differently across various domains. For instance, on the ECL dataset, removing it caused a 5% increase in MSE, indicating that AKF enables refined processing of dynamic data across heterogeneous high-dimensional variables. However, in the “AKF-only” combination ablation experiment, model performance declined—by approximately 3.38% on the Exchange dataset. This indicates that the AKF module, functioning as a preprocessing and denoising component, requires synergistic interaction with other modeling components. This further corroborates that the design of the other two components explicitly considered the role of AKF and incorporated targeted design considerations.

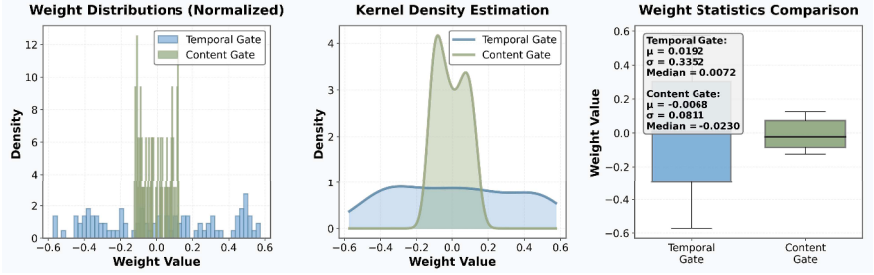
Secondly, the ablation of MSED demonstrates its ability to capture time dependencies across different scales. Given the indispensable role of embedding layers in Transformer-based models, the ablation experiment for MSED involved replacing it with a traditional single-scale embedding layer (SSE). The performance of the replaced model deteriorated significantly across all datasets, with the average MSE increasing by 4.97%. This impact was particularly pronounced on datasets exhibiting pronounced periodicity and trends. For instance, removing this module resulted in an MSE increase of 6.99% on the ETTh1 dataset and 5.83% on the Weather dataset. More critically, the performance gap widened dramatically with increasing forecast lengths (e.g., 720-step predicted length), highlighting MSED’s key advantage in modeling complex long-term patterns through hierarchical scale-aware representations. Additionally, in combined ablation experiments, the “only MSED”



(a)Visualization of TempGate



(b) Visualization of ContentGate



(c) Dual-Gate Comparison

Fig. 8. Visualization of DGFA dual-gate mechanism.

model variant showed no significant performance degradation across many datasets (particularly with only a 1.2% MSE reduction on ETTh1), further confirming this component's central role in multiscale modeling and fusion. To further isolate the effect of attention-based fusion, we replace MSED's attention with learnable scalar weights (LSW). As shown in Table 5, LSW consistently underperforms AGformer across all datasets, with average MSE increases of 4.6% on ETTh1, 9.4% on ECL, 2.7% on Weather, and 3.4% on Exchange, confirming that attention-based inter-scale interaction provides essential cross-resolution dependencies beyond static weighting.

Finally, the DGFA module demonstrates certain value when processing data with implicit spatial or structural correlations. Ablation experiments for this module, as shown in Tables 5 and 6, are categorized into four scenarios: retaining only DGFA (combined ablation of AKF and MSED), replacing the entire module with self-attention (Abl-1), adopting a gated self-attention mechanism (Abl-2), and applying only the fusion attention mechanism (Abl-3). Case (Abl-1) directly replaces it with standard self-attention (SA) while simultaneously removing the gating and fusion mechanisms, resulting in a comprehensive decline in model performance: MSE increases by 5% on ETTh1, 2.32% on Exchange, and 6.71% on Weather.

Furthermore, the gating and fusion mechanisms require in-depth analysis by combining the results from Case (Abl-2) and Case (Abl-3). Observing the experiments in Case (Abl-2) reveals that relying solely on the gating mechanism without effective fusion leads to over-filtering, causing a significant decline in model performance—for instance, a 15%-20% drop on the Exchange dataset. Similarly, results from Case (Abl-3) demonstrate that the model's predictive accuracy also drops substantially when no interaction exists between two inputs: ETTh1 increases by 30%-40% and Weather increases by 20%-25%. Therefore, this indicates that the gating mechanism and fusion mechanism of DGFA are complementary in design and function synergistically. It is noteworthy that when "only DGFA" in combined ablation, the performance resembles that of "only AKF," with the model's standalone modeling capability declining. This confirms the critical role of AKF and DGFA in collaborative division of labor. This effect is particularly pronounced on the Weather dataset, indicating that these two components effectively uncover complex dependencies in datasets with heterogeneous temporal dynamics.

In summary, the superiority of this model stems from the distinct yet synergistic roles of its three components. The absence of any one or two components would result in degraded model performance. MSED forms the foundational framework for multi-scale feature extraction, upon which AKF implements adaptive fine filtering. Meanwhile, DGFA captures the relationship between long-term and short-term information through the coordinated action of its gating and fusion mechanisms.

Table 5
A results of individual ablation and combined ablation.

Datasets		ETTh1		ECL		Weather		Exchange	
Methods	len	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
AGformer	96	**0.372**	**0.401**	**0.131**	**0.225**	**0.147**	**0.196**	**0.083**	**0.202**
	192	**0.406**	**0.422**	**0.148**	**0.241**	**0.190**	**0.240**	**0.176**	**0.300**
	336	**0.428**	**0.440**	**0.163**	**0.258**	**0.243**	**0.280**	**0.327**	**0.414**
	720	**0.455**	**0.476**	**0.201**	**0.290**	**0.311**	**0.329**	**0.836**	**0.688**
	Avg	**0.415**	**0.434**	**0.160**	**0.253**	**0.223**	**0.262**	**0.355**	**0.401**
w/o AKF	96	0.381	0.403	0.136	0.229	0.150	0.201	0.087	0.205
	192	0.415	0.424	0.154	0.245	0.194	0.243	0.179	0.302
	336	0.434	0.446	0.171	0.263	0.251	0.287	0.334	0.420
	720	0.462	0.480	0.210	0.296	0.328	0.343	0.851	0.696
	Avg	0.423	0.438	0.168	0.258	0.231	0.268	0.363	0.406
MSED → SSE	96	0.396	0.423	0.134	0.229	0.161	0.213	0.089	0.211
	192	0.426	0.442	0.151	0.245	0.204	0.253	0.186	0.308
	336	0.449	0.459	0.168	0.262	0.257	0.292	0.328	0.415
	720	0.504	0.508	0.207	0.296	0.321	0.340	0.871	0.703
	Avg	0.444	0.458	0.165	0.258	0.236	0.274	0.369	0.409
MSED → LSW	96	0.392	0.415	0.142	0.239	0.152	0.201	0.091	0.211
	192	0.428	0.437	0.160	0.257	0.195	0.243	0.189	0.318
	336	0.447	0.451	0.179	0.275	0.247	0.284	0.345	0.429
	720	0.464	0.481	0.219	0.309	0.322	0.337	0.844	0.701
	Avg	0.434	0.446	0.175	0.270	0.229	0.266	0.367	0.415
only AKF	96	0.390	0.405	0.159	0.251	0.186	0.227	0.088	0.208
	192	0.442	0.435	0.170	0.261	0.233	0.266	0.181	0.303
	336	0.481	0.457	0.186	0.277	0.287	0.303	0.334	0.419
	720	0.485	0.479	0.223	0.307	0.355	0.341	0.863	0.702
	Avg	0.450	0.444	0.185	0.274	0.265	0.284	0.367	0.408
only MSED	96	0.402	0.461	0.169	0.246	0.168	0.219	0.089	0.217
	192	0.449	0.470	0.177	0.258	0.216	0.254	0.186	0.303
	336	0.447	0.486	0.199	0.273	0.264	0.301	0.342	0.418
	720	0.451	0.508	0.225	0.795	0.333	0.325	0.916	0.696
	Avg	0.420	0.481	0.183	0.381	0.241	0.273	0.383	0.409
only DGFA	96	0.389	0.404	0.167	0.255	0.184	0.225	0.087	0.207
	192	0.439	0.435	0.172	0.261	0.234	0.266	0.180	0.304
	336	0.478	0.457	0.183	0.273	0.289	0.305	0.324	0.413
	720	0.485	0.481	0.221	0.303	0.361	0.351	0.871	0.706
	Avg	0.447	0.4448	0.186	0.273	0.267	0.287	0.366	0.408

[a] The best results are highlighted in bold and the second best results are in underlined.
[b] SSE represents *Single-Scale Embedding*.
[c] LSW represents replacing the MSED attention-based fusion scheme with a learnable scalar weight fusion scheme.

Table 6
Ablation study results of DGFA.

Design	Gate	Fusion	len	ETTh1		Exchange		Weather	
				MSE	MAE	MSE	MAE	MSE	MAE
AGformer	✓	✓	96	**0.372**	**0.401**	**0.083**	**0.202**	**0.147**	**0.197**
			192	**0.406**	**0.422**	**0.176**	**0.300**	**0.192**	**0.241**
			336	**0.428**	**0.440**	**0.327**	**0.414**	**0.244**	**0.280**
			720	**0.455**	**0.476**	**0.836**	**0.688**	**0.311**	**0.329**
Abl-1	w/o	w/o	96	0.388	0.415	0.085	0.205	0.163	0.214
			192	0.424	0.441	0.178	0.304	0.207	0.254
			336	0.443	0.456	0.340	0.423	0.257	0.293
			720	0.489	0.497	0.852	0.698	0.327	0.342
Abl-2	✓	w/o	96	0.709	0.574	0.137	0.268	0.273	0.323
			192	0.713	0.583	0.239	0.354	0.297	0.338
			336	0.715	0.589	0.397	0.462	0.325	0.354
			720	0.731	0.617	0.961	0.746	0.369	0.383
Abl-3	w/o	✓	96	0.530	0.494	0.135	0.266	0.201	0.257
			192	0.562	0.516	0.236	0.353	0.254	0.297
			336	0.568	0.524	0.391	0.458	0.301	0.330
			720	0.655	0.582	0.973	0.752	0.357	0.369

Table 7
Model complexity comparison.

Model	Time Complexity	Space Complexity
AGformer	$O(C^2)$	$O(C^2)$
Crossformer	$O(L \times \text{seg_len})$	$O(\text{seg_num}^2)$
FEDformer	$O(L \log L)$	$O(L)$
DLinear	$O(L)$	$O(1)$
iTransformer	$O(C^2)$	$O(C^2)$
Nonstationary	$O(L^2)$	$O(L^2)$
TimeMixer	$O(L \log L)$	$O(L)$
TimesNet	$O(L \log L)$	$O(L)$

4.5. Model complexity and computational cost

(1) Complexity Analysis

In practical applications of time series forecasting, model complexity and computational efficiency are key factors determining the feasibility of deployment. This section provides a detailed analysis of the computational complexity of the AGformer model for each component, comparing it with six mainstream models: DLinear, TimesNet, iTransformer , Crossformer, TimeMixer, and Nonstationary. Among these, DLinear is an MLP-based model, making AGformer more comparable to other Transformer-based models. The complexity of the AKF module is as follows.

$$C_{\text{AKF}} = \mathcal{O}(B \times L \times C \times w) \tag{26}$$

where B is the batch size, L is the sequence length, C is the number of variables, and w is the filter window size. The complexity of the MSED module is as follows.

$$C_{\text{MSED}} = \mathcal{O}(B \times L \times d_{\text{model}} \times s) \tag{27}$$

where d_{model} denotes the embedding dimension, and s represents the multiscale quantity. The complexity of the DGFA module is as follows.

$$C_{\text{DGFA}} = \mathcal{O}(\underbrace{B \times C \times d_{\text{model}}^2}_{\text{Linear projection}} + \underbrace{B \times d_{\text{model}} \times C \times n}_{\text{TempGate}} + \underbrace{B \times d_{\text{model}} \times C^2}_{\text{ContentGateand + Attn}}) \tag{28}$$

where n is the size of the convolution kernel. Additionally, considering the computational complexity of the FFN layer, projection layer, and other foundational components, the overall complexity of AGformer is concisely represented as follows.

$$C_{\text{AGformer}} = \mathcal{O}\big(B \times \big[L \times (C \times w + d_{\text{model}} \times s) + E \times (C \times d_{\text{model}}^2 + d_{\text{model}} \times C^2 + d_{\text{model}} \times C \times n)\big]\big) \tag{29}$$

Therefore, the dominant term in AGformer's time complexity is proportional to the square of the number of variables C, with linear scaling in the time dimension and quadratic scaling in the variable dimension. By sequentially calculating the time complexity and space complexity of the other models, we obtain Table 7.

As shown in Table 7, DLinear exhibits the lowest complexity, with both time and space complexity being linear, making it the most computationally efficient. However, linear models may struggle to capture complex temporal dependencies and nonlinear patterns. Among Transformer-based models, Nonstationary exhibits $O(L^2)$ time and space complexity, limiting its ability to process long sequences. FEDformer and TimeMixer, and TimesNet reduce temporal complexity to $O(L \log L)$ through decomposition operations. In contrast, the proposed AGformer and iTransformer both shift attention from the temporal dimension to the variable dimension, resulting in temporal complexity proportional to the square of the number of variables C and linearly related to sequence length L. Therefore, if the sequence length L is large and the number of variables C is small, AGformer and iTransformer hold an advantage.

(2) Computational Cost Analysis

In this section, we evaluate each model by analyzing its parameters, GPU memory usage, computational complexity (FLOPs), inference

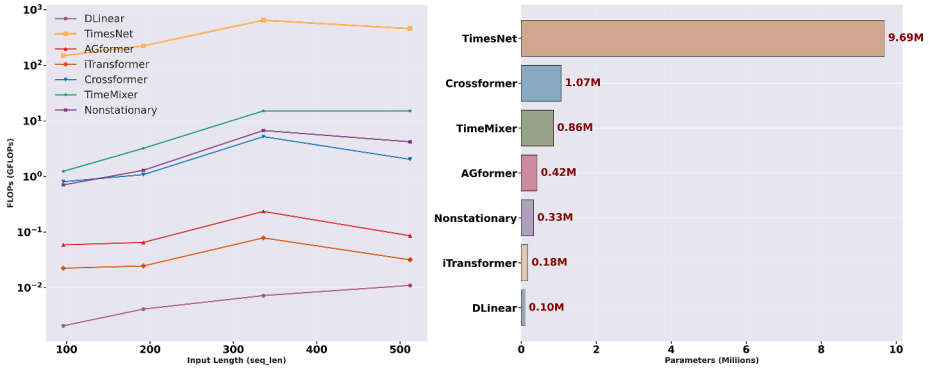


Fig. 9. Comparison of FLOPs and parameters calculation results.

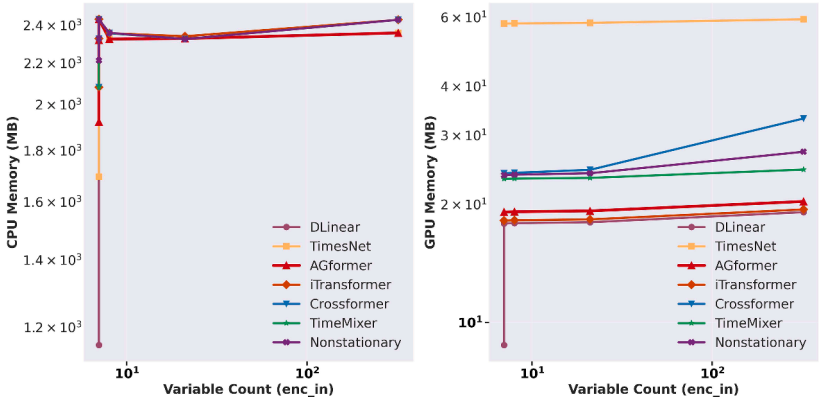


Fig. 10. CPU/GPU memory usage comparison results.

time, and training time. All metrics were evaluated based on the following configuration: input sequence lengths of (96, 192, 336, 512), a prediction length of 96, and (7, 21, 321) variables.

As shown in Fig. 9, AGformer exhibits a slow increase in FLOPs with sequence length and maintains a relatively low overall value. Among all models, it ranks among the top three in computational efficiency, comparable to Dlinear and iTransformer, while outperforming most other Transformer-based models. Moreover, AGformer's parameter count (0.42M) is moderately low, only 4.2 times that of the baseline linear model DLinear, significantly lower than TimesNet (9.69M) and Crossformer (1.07M). This demonstrates that AGformer maintains a relatively compact parameter design while adopting a multi-component architecture, showcasing exceptional parameter efficiency.

Further analysis of CPU/GPU memory utilization reveals the results shown in Fig. 10. AGformer exhibits stable CPU memory consumption comparable to other mainstream Transformer-based models. Regarding GPU memory, AGformer demonstrates strong performance and predictability. Its memory usage remains low even with multi-variable inputs, indicating excellent scalability.

In terms of training time, AGformer took 30.70ms to train. While still slower than the lightest models DLinear (7.04ms) and TimeMixer (17.26ms), it demonstrated advantages over other complex models: it was 71.5% faster than TimesNet and 44.5% faster than Crossformer, but 16.6% slower than iTransformer. The efficiency ratio of AGformer is 0.066, placing it at a moderate level and indicating a reasonable balance between training and inference. This ratio is comparable to Nonstationary and outperforms Crossformer.

(3) Positioning Relative to Time-Series Foundation Models.

Table 8 further benchmarks AGformer against large-scale time-series foundation models (TSFMs). The results reveal a significant disparity in efficiency: AGformer employs only 0.42M parameters (approximately 1/300 of the parameter count of the aforementioned TSFMs) and achieves an inference time of 2.03ms on an RTX 4090D GPU. In contrast, even when estimated on the modern RTX 5090 GPU, the inference latencies for Sundial and Chronos-2 remain as high as 17.5ms and 12.01ms, respectively. This gap is particularly striking when considering that the CPU inference time originally reported for Sundial was as high as 249ms. These quantitative comparisons demonstrate that, while TSFMs possess powerful zero-shot generalization capabilities, their massive parameter scales and high inference overheads render them less suitable for deployment scenarios characterized by resource constraints or low-latency requirements. Conversely, AGformer is designed as a lightweight, domain-adaptive solution that prioritizes efficiency and interpretability, thereby complementing the cross-domain versatility of TSFMs.

Table 8
Comparison of model training time and inference time.

Model	Training Time (ms)	Inference Time (ms)	Parameters	efficiency ratio
AGformer	30.7	2.03	0.42M	0.066
TimesNet	102.29	3.41	9.69M	0.033
Crossformer	55.3	4.61	1.07M	0.083
iTransformer	26.33	0.87	3.38M	0.033
TimeMixer	17.26	0.58	0.74M	0.034
DLinear	7.04	0.26	0.065M	0.037
Sundial	N/A (pre-trained)	17.5ms	128M	—
Chronos-2	N/A (pre-trained)	12.01ms	120M	—

"–" indicates that the efficiency ratio is not applicable for pre-trained foundation models, as they do not require per-dataset training.

4.6. Robustness analysis

(1) Cross-Domain Generalization Validation

To validate the cross-domain generalization capability of AGformer, we introduce the PJM Interconnection dataset, which differs substan-

Table 9
Cross-domain generalization results.

Models		AGformer		iTransformer		FEDformer		Crossformer		TimeMixer		TimesNet		DLinear		Stationary	
Datasets	len	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.372**	**0.401**	0.386	0.405	0.379	0.419	0.411	0.435	**0.372**	**0.401**	0.389	0.412	0.379	0.403	0.591	0.524
	192	**0.406**	0.422	0.424	0.440	0.420	0.444	0.409	0.438	0.413	0.43	0.440	0.443	0.408	**0.419**	0.615	0.540
	336	**0.428**	**0.440**	0.449	0.460	0.458	0.466	0.433	0.457	0.438	0.45	0.523	0.487	0.440	**0.440**	0.632	0.551
	720	**0.455**	**0.476**	0.495	0.487	0.474	0.488	0.501	0.514	0.486	0.484	0.521	0.495	0.471	0.493	0.828	0.658
	Avg	**0.415**	**0.434**	0.439	0.448	0.433	0.454	0.439	0.461	0.427	0.441	0.468	0.459	0.425	0.439	0.667	0.568
ECL	96	**0.131**	**0.225**	0.134	0.230	0.191	0.305	0.134	0.231	0.153	0.256	0.169	0.271	0.140	0.237	0.171	0.274
	192	0.148	**0.241**	0.154	0.250	0.203	0.316	**0.146**	0.243	0.168	0.269	0.180	0.280	0.154	0.251	0.180	0.283
	336	**0.163**	**0.258**	0.169	0.265	0.221	0.333	0.165	0.264	0.189	0.291	0.204	0.304	0.169	0.268	0.204	0.305
	720	0.201	0.290	**0.194**	**0.288**	0.259	0.364	0.237	0.314	0.228	0.320	0.205	0.304	0.204	0.301	0.221	0.319
	Avg	**0.160**	**0.253**	0.163	0.258	0.219	0.330	0.170	0.263	0.185	0.284	0.189	0.290	0.167	0.264	0.194	0.295
PJM	96	**0.301**	**0.393**	0.318	0.406	0.310	0.408	0.319	0.409	0.315	0.404	0.309	0.404	0.362	0.445	0.359	0.437
	192	0.401	0.462	0.445	0.486	0.408	0.477	**0.373**	**0.454**	0.422	0.472	0.408	0.474	0.439	0.491	0.442	0.500
	336	**0.473**	**0.509**	0.519	0.529	0.493	0.528	0.513	0.525	0.495	0.517	0.475	0.511	0.489	0.519	0.493	0.525
	720	0.554	0.550	0.581	0.561	0.598	0.588	0.573	0.563	**0.553**	**0.548**	0.554	0.557	0.562	0.563	0.522	0.539
	Avg	**0.432**	**0.478**	0.466	0.496	0.452	0.500	0.445	0.488	0.446	0.485	0.437	0.487	0.463	0.505	0.454	0.500

The best results are highlighted in bold and the second best results are in underlined.

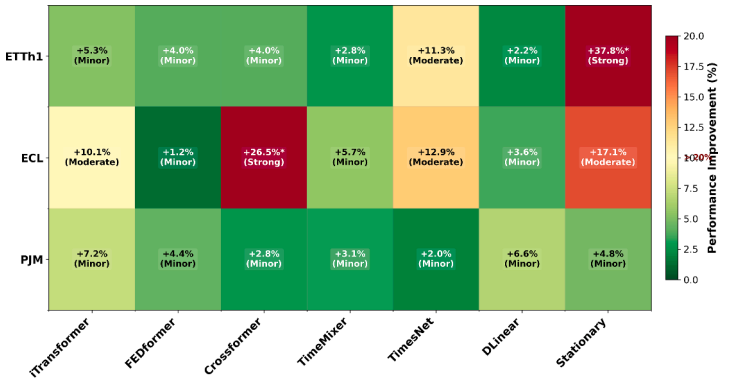


Fig. 11. AGformer performance enhancement heatmap.

tially from previously evaluated datasets in terms of variable count (dozens of regional sub-loads), spatial coupling structure (multi-regional interconnection), and temporal dynamics (aggregated grid-level patterns). This diversity—spanning device-level sensors (ETT), user-level meters (ECL), and grid-level regional aggregates (PJM)—provides a rigorous testbed for assessing whether the model's inductive biases transfer across fundamentally different data regimes.

Specifically, we constructed a multivariate forecasting task on PJM: using historical loads from all other sub-regions to predict future values for the core region PJME. Table 9 and Fig. 11 demonstrate that AGformer consistently achieves the best average performance across all three datasets. The performance enhancement heatmap in Fig. 11 reveals that improvement margins vary meaningfully across datasets: AGformer yields relatively stronger gains on ECL (11% average MSE improvement) than on ETTh1 (9.63%) and PJM (4.41%), which can be attributed to the module-level inductive biases of AGformer. On ECL—a high-dimensional dataset with 321 heterogeneous user-level variables—the AKF's per-variable adaptive filtering provides significant benefits by accommodating the wide spectral diversity across users. On ETTh1, the MSED's multi-scale attention fusion captures the well-structured diurnal and weekly periodicity effectively. The comparatively smaller gain on PJM reflects that multi-regional aggregate loads exhibit smoother, more homogeneous spectral profiles, which reduce the marginal contribution of adaptive per-variable filtering while the DGFA's short-term fluctuation modeling still provides measurable advantage over baselines that discard such signals.

These results demonstrate that AGformer's three modules provide complementary, transferable inductive biases: AKF's adaptive spectral parameterization, DGFA's heterogeneous information fusion, and MSED's attention-based multi-scale aggregation collectively enable consistent performance across datasets that vary substantially in variable count, temporal resolution, and spatial coupling structure.

(2) Robustness to Abrupt Structural Breaks

As motivated in Section 1, short-term fluctuations arising from abrupt structural breaks and sudden load transitions are systematically suppressed by standard denoising, yet they encode critical information about system state changes. To directly validate AGformer's robustness to such inputs, we identify two representative event windows using a combination of manual annotation and sequence rate-of-change detection. As shown in Fig. 12, the HUFL channel of ETTh1 exhibits a sharp structural break within the time window [10470, 10480], representing an abrupt system-level signal transition. Channel 1 of the Electricity dataset shows a sudden signal drop within [19650, 19660], representing a rapid change in the underlying generative process. Both scenarios correspond precisely to the class of short-term fluctuations that DGFA is designed to preserve via its dual-gate cross-attention mechanism.

AGformer maintains stable, distortion-free predictions under both structural break scenarios without exhibiting the prediction collapse

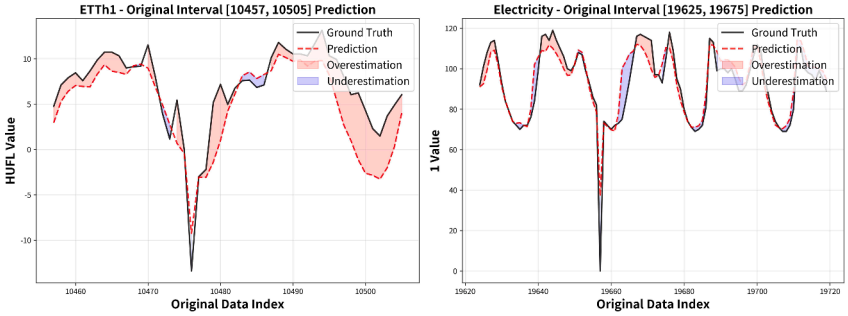


Fig. 12. Prediction results under extreme input anomalies.

observed in models that over-smooth such transitions. Minor deviations occur immediately following the event onset, but the model rapidly recovers, confirming that DGFA's selective reintroduction of short-term fluctuation signals—identified as a key architectural gap in Section 1—translates into measurable robustness on real-world non-stationary inputs.

(3) Limitations and Future Work

Although AGformer has strong empirical performance, it still has some limitations. First, the AKF module is based on global FFT and cannot adequately capture the transient spectral characteristics of highly non-stationary signals. In the future, alternative time-frequency representations such as short-time Fourier transform or wavelet transform can be explored. Second, DGFA uses original data as a query, and extreme outliers or noise may be amplified by the gating mechanism, so robust preprocessing or outlier-aware gating design needs to be introduced. Third, the model complexity increases quadratically with the number of variables C. When $C > 500$, the computational overhead is significant. Dimensionality reduction or variable grouping can be used to improve scalability. Fourth, AGformer only supports independent training on a per-dataset basis and lacks online distribution adaptability. Combining incremental learning or meta-learning may be a feasible improvement direction. The above limitations point out the path for subsequent research, including hybrid spectral decomposition, robust gating mechanisms and efficient attention variants in high-dimensional scenes.

5. Conclusion

This paper proposes AGformer, a multivariate time series forecasting architecture that addresses three persistent neural network design challenges through three synergistic modules. Experiments on nine benchmark datasets across electricity, weather, finance, and transportation domains demonstrate that: 1) the frequency-domain based AKF can dynamically optimize the per-variable filtering strategy by learning from spectral features (mean energy, spectral entropy, energy concentration), enabling the model to adapt to heterogeneous channel characteristics rather than applying a global hyperparameter; 2) the DGFA synergistically integrates long-term trend representations with short-term fluctuation signals through dual gating, improving robustness to abrupt structural breaks that standard denoising would suppress; 3) the MSED captures multi-resolution temporal dependencies through attention-based inter-scale fusion, outperforming simplistic concatenation or static weighted summation strategies.

From a neural network design perspective, the core contributions of AGformer—adaptive spectral parameterization, dual-pathway gating for heterogeneous information fusion, and attention-based multi-scale aggregation—are general principles applicable beyond the forecasting benchmarks evaluated here. The AKF's learnable frequency selection mechanism is directly applicable to any sequence modeling domain where channels exhibit heterogeneous spectral complexity, such as biomedical signal analysis or acoustic modeling. The DGFA's cross-attention gating framework offers a principled approach to fusing representations of different information granularities, a challenge that arises in natural language processing (e.g., integrating local syntax and global semantics) and multimodal learning. Future research will systematically compare the performance of AGformer and TSFM under zero-shot and fine-tuning settings and explore integrating TSFM's pre-trained representations into AGformer to enhance cross-domain adaptability; additionally, emerging backbone architectures—such as state-space models and linear attention—will be incorporated to further compress the models for deployment in resource-constrained environments.

CRediT authorship contribution statement

Shu-Xin Xu: Writing – original draft, Visualization, Methodology; **Jiang-Wen Xiao:** Writing – review & editing, Funding acquisition; **Peng-Cheng Li:** Writing – review & editing; **Tian-Hao Lu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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References

Ansari, A. F., Shchur, O., Küken, J., Auer, A., Han, B., Mercado, P., Rangapuram, S. S., Shen, H., Stella, L., Zhang, X., Goswami, M., Kapoor, S., Maddix, D. C., Guerron, P., Hu, T., Yin, J., Erickson, N., Mutalik Desai, P., Wang, H., Rangwala, H., Karypis, G., Wang, Y., & Bohlke-Schneider, M. (2025). Chronos-2: From univariate to universal forecasting. arXiv:2510.15821.

Bai, S., Kolter, J. Z., & Koltun, V. (2018). An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. https://doi.org/10.48550/arXiv.1803.01271

Chen, Y., Liu, S., Yang, J., Jing, H., Zhao, W., & Yang, G. (2024). A joint time-frequency domain transformer for multivariate time series forecasting. *Neural Networks*, *176*, 106334.

Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)* (pp. 1724–1734).

Das, A., Kong, W., Sen, R., & Zhou, Y. (2024). A decoder-only foundation model for time-series forecasting. In *International conference on machine learning (ICML)*.

Dauphin, Y. N., Fan, A., Auli, M., & Grangier, D. (2017). Language modeling with gated convolutional networks. In *Proceedings of the 34th international conference on machine learning (ICML)* (pp. 933–941). PMLR.

Eldele, E., Ragab, M., Chen, Z., Wu, M., & Li, X. (2024). TSLANet: Rethinking transformers for time series representation learning. https://doi.org/10.48550/arXiv.2404.08472

Fu, K., Li, H., & Shi, X. (2024). CTF-former: A novel simplified multi-task learning strategy for simultaneous multivariate chaotic time series prediction. *Neural Networks*, *174*, 106234.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep learning. MIT Press. http://www.deeplearningbook.org.

Goswami, M., Szafer, K., Choudhry, A., Cai, Y., Li, S., & Dubrawski, A. (2024). Moment: A family of open time-series foundation models. In *International conference on machine learning (ICML)*.

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, *9*(8), 1735–1780.

Jin, M., Wang, S., Ma, L., Chu, Z., Zhang, J. Y., Shi, X., Chen, P. Y., Liang, Y., Li, Y. F., Pan, S., & Wen, Q. (2024). Time-LLM: Time series forecasting by reprogramming large language models. In *International conference on learning representations (ICLR)*.

Li, A., Li, Y., Xu, Y., Li, X., & Zhang, C. (2024a). Multi-scale convolution enhanced transformer for multivariate long-term time series forecasting. *Neural Networks*, *180*, 106745.

Li, B. et al. (2025). Integrated codec decomposed transformer for long-term series forecasting. *Neural Networks*, *188*, 107484.

Li, D., Fan, S., Ma, W., Li, S., Zhao, Z., & Gao, Z. (2024b). Ultra-short-term load forecasting method based on improved iTransformer of decomposition-reconstruction. In *2024 IEEE 8th conference on energy internet and energy system integration (EI2)* (pp. 225–230). IEEE.

Li, S., Jin, X., Xuan, Y., Zhou, X., Chen, W., Wang, Y. X., & Yan, X. (2019). Enhancing the locality and breaking the memory bottleneck of transformer on time series forecasting. *Advances in Neural Information Processing Systems*, *32*, 5243–5253.

Lin, S., Chen, H., Wu, H., Qiu, C., & Lin, W. (2025). Temporal query network for efficient multivariate time series forecasting. In *International conference on machine learning (ICML)*.

Liu, M., Xia, C., Xia, Y., Deng, S., & Wang, Y. (2025a). TDCN: A novel temporal depthwise convolutional network for short-term load forecasting. *International Journal of Electrical Power & Energy Systems*, *165*, 110512.

Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L., & Long, M. (2023). iTransformer: Inverted transformers are effective for time series forecasting. https://doi.org/10.48550/arXiv.2310.06625

Liu, Y., Qin, G., Shi, Z., Chen, Z., Yang, C., Huang, X., Wang, J., & Long, M. (2025b). Sundial: A family of highly capable time series foundation models. In *International conference on machine learning (ICML)*.

Liu, Y., Wu, H., Wang, J., & Long, M. (2022). Non-stationary transformers: Exploring the stationarity in time series forecasting. *Advances in Neural Information Processing Systems*, *35*, 9881–9893.

Pentsos, V., Tragoudas, S., Wibbenmeyer, J., & Khdeer, N. (2025). A hybrid LSTM-transformer model for power load forecasting. *IEEE Transactions on Smart Grid*, *16*(3), 2624–2634. https://doi.org/10.1109/TSG.2025.3535407

Rahman, R., Kumar, N., & Nguyen, D. C. (2025). Electrical load forecasting in smart grid: A personalized federated learning approach. In *2025 IEEE 22nd consumer communications & networking conference (CCNC)* (pp. 1–2).

Sun, J., Sun, Z., Chen, Z., Dong, M., Wang, X., Chen, C., Zheng, H., & Zhao, X. (2025). MSA-LR: Enhancing multi-scale temporal dynamics in multivariate time series forecasting with low-rank self-attention. *Neural Networks*, *194*, (p. 108150).

Tong, H., & Liu, J. (2025). MFformer: An improved transformer-based multi-frequency feature aggregation model for electricity load forecasting. *Electric Power Systems Research*, *243*, 111492.

Trindade, A. (2015). ElectricityLoadDiagrams20112014. UCI Machine Learning Repository. https://doi.org/10.24432/C58C86

van den Oord, A., Dieleman, S., Zen, H., Simonyan, K., Vinyals, O., Graves, A., Kalchbrenner, N., Senior, A., & Kavukcuoglu, K. (2016). WaveNet: A generative model for raw audio. https://doi.org/10.48550/arXiv.1609.03499

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Advances in neural information processing systems*. (*vol. 30*).

Wan, J., Xia, N., Yin, Y., Pan, X., Hu, J., & Yi, J. (2024). TCDformer: A transformer framework for non-stationary time series forecasting based on trend and change-point detection. *Neural Networks*, *173*, 106196.

Wang, S., Wu, H., Shi, X., Hu, T., Luo, H., Ma, L., Zhang, J. Y., & Zhou, J. (2024a). TimeMixer: Decomposable multiscale mixing for time series forecasting. https://doi.org/10.48550/arXiv.2405.14616

Wang, Z., Kong, F., Feng, S., Wang, M., Zhao, H., Wang, D., & Zhang, Y. (2024b). Is mamba effective for time series forecasting? *Neurocomputing*, *619*, (p. 129178).

Wu, H., Hu, T., Liu, Y., Zhou, H., Wang, J., & Long, M. (2022). TimesNet: Temporal 2D-variation modeling for general time series analysis. https://doi.org/10.48550/arXiv.2210.02186

Wu, H., Xu, J., Wang, J., & Long, M. (2021). Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. In *Advances in neural information processing systems* (pp. 22419–22430). Curran Associates, Inc. (*vol. 34*).

Xiao, J. W., Liu, P., Fang, H., Liu, X. K., & Wang, Y. W. (2023). Short-term residential load forecasting with baseline-refinement profiles and bi-attention mechanism. *IEEE Transactions on Smart Grid*, *15*(1), 1052–1062.

Xu, X., Chen, C., Liang, Y., Huang, B., Bai, G., Zhao, L., & Shu, K. (2025). SST: Multi-scale hybrid mamba-transformer experts for time series forecasting. In *Proceedings of the 34th ACM international conference on information and knowledge management (CIKM)* (pp. 3655–3665).

Yang, X., Li, H., Huang, X., & Feng, X. (2024). FEDAF: Frequency enhanced decomposed attention free transformer for long time series forecasting. *Neural Computing and Applications*, *36*(26), 16271–16288.

Zeng, A., Chen, M., Zhang, L., & Xu, Q. (2023). Are transformers effective for time series forecasting? In *Proceedings of the AAAI conference on artificial intelligence* (pp. 11121–11128). (*vol. 37*).

Zhang, S., Yi, H., Zeng, F., Fu, Z., An, D. et al. (2025). Diffmixer: A prediction model based on mixing different frequency features. *Neural Networks*, *194*, (p. 108188).

Zhang, T., Zhang, Y., Cao, W., Bian, J., Yi, X., Zheng, S., & Li, J. (2022). Less is more: Fast multivariate time series forecasting with light sampling-oriented MLP structures. https://doi.org/10.48550/arXiv.2207.01186

Zhang, Y., & Yan, J. (2023). Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting. In *The eleventh international conference on learning representations*.

Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., & Zhang, W. (2020). Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Aaai conference on artificial intelligence*.

Zhou, T., Ma, Z., Wen, Q., Wang, X., Sun, L., & Jin, R. (2022). FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting. In *International conference on machine learning* (pp. 27268–27286). PMLR.

Zhou, Z., Lyu, G., Huang, Y., Wang, Z., Jia, Z., & Yang, Z. (2024). SDformer: Transformer with spectral filter and dynamic attention for multivariate time series long-term forecasting. In *Proceedings of the thirty-third international joint conference on artificial intelligence (IJCAI-24)* (pp. 3–9).